Rational Dynamics Notes

Adrián Baldellou Aguirre

June 13, 2026

Prerequisites

From Complex Analysis

- d-fold map: We say f is a d-fold map of V onto W if, for every $w \in W$, the equation $f(z) = w$ has exactly d solutions in V counting multiple solutions (by their multiplicity).
- Maximum Modulus Theorem.
- Schwarz's Lemma.
- Rouché's Theorem.
- Hurwitz's Theorem: If a sequence of univalent analytic maps f_n on a domain D converges uniformly to f, then f is either constant or univalent in D.

From Topology

- Metric spaces: Uniform continuity.
- Compactness.
- Connectedness.

Contents

Chapter 1

Examples

In this chapter we introduce some of the main ideas in iteration theory by discussing a variety of simple examples. The discussions involve only elementary mathematics, and our sole objective is to illustrate and stress those features that will be met in a general context later.

1.1 Introduction

Proposition 1.1 ($z_n \xrightarrow[n\to\infty]{} w \implies w$ is a fixed point)**.** *Let $z_0 \in \mathbb{C}_\infty$, R be a rational function in $\mathbb{C}_\infty$, and $z_n = R^n(z_0)$. Then if z_n converges when n goes to infinity, $\lim_{n\to\infty} z_n = w$ is a fixed point of R.*

Proof. Since R is continuous (everywhere where it is defined, i.e., the denominator is not zero) because it is a rational function, the following is true:

$$w = \lim_{n\to\infty} z_n = \lim_{n\to\infty} R(z_{n-1}) = R\left(\lim_{n\to\infty} z_{n-1}\right) = R(w).$$

Then, $w = R(w)$, so it is a fixed point. □

Intuition: Let's see why the derivative can work for determining if a point is attracting or repelling. Since $|R(z) - \zeta| = |R(z) - R(\zeta)| \simeq |R'(\zeta)||z - \zeta|$ if z is close enough to ζ, we have that if $|R'(\zeta)| < 1$, $z_n \to \zeta$, and if $|R'(\zeta)| > 1$, z_n will be initially repelled from ζ. (A rigorous contradiction proof follows in the notes ensuring z_{n+1} cannot arbitrarily stay closer than z_n if repelled).

1.2 Iteration of Möbius Transformations

Definition 1.1 (Möbius Transformation)**.** A *Möbius transformation* is a rational function $R : \mathbb{C}_\infty \to \mathbb{C}_\infty$ of the form

$$R(z) = \frac{az+b}{cz+d},$$

where $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$.

Theorem 1.2 (Classification of Möbius Transformations)**.** *Given* R *a Möbius transformation, then either it has:*

1. *One fixed point,* ζ*, and for every* $z \in \mathbb{C}$*,* $R^n(z) \to \zeta$*.*
2. *Two fixed points,* ζ_1 *and* ζ_2*, and for every* $z \in \mathbb{C}_\infty$ *either:*
   (a) $R^n(z) \to \zeta_1$
   (b) $R^n(z) \to \zeta_2$
   (c) $R^n(z)$ *moves cyclically over a finite set of points.*
   (d) $\{R^n(z) : n \in \mathbb{N}_0\}$ *forms a dense subset of a circle.*

Proof. First, observe that R cannot have 0 fixed points. Solving $R(z) = z$:

$$\frac{az+b}{cz+d} = z \implies cz^2 + (d-a)z - b = 0$$

This is a quadratic equation, so it always has 1 or 2 solutions in $\mathbb{C}_\infty$ (if $c = 0$, ∞ is a fixed point).

Case 1: One fixed point. Suppose the fixed point is ∞. Then $R(z) = z + \beta$ for some $\beta \in \mathbb{C}$:

$$R(\infty) = \infty \implies c = 0 \implies R(z) = \frac{a}{d}z + \frac{b}{d} = \alpha z + \beta.$$

We also want R to have no other fixed point:

$$R(z) = z \iff \alpha z + \beta = z \iff (\alpha - 1)z + \beta = 0,$$

hence β has to be $\neq 0$ (otherwise $z = 0$ is a fixed point) and $\alpha = 1$ (otherwise $z = \beta/(\alpha - 1)$ is a fixed point).

If R has a general fixed point ζ, we can define the Möbius transformation S by using the conjugation $g(z) = \frac{1}{z-\zeta}$ and $S = g \circ R \circ g^{-1}$. Since conjugation preserves the number of fixed points, S has exactly one fixed point at ∞, meaning S is a translation $S(z) = z + \beta$. Thus, $S^n(z) \to \infty$, which implies $R^n(z) \to \zeta$.

Case 2: Two fixed points. Suppose the fixed points are 0 and ∞ (it can be shown similarly to the previous case). Then $R(z) = kz$ for some $k \in \mathbb{C} \setminus \{0, 1\}$. Analyzing $R^n(z) = k^n z$:

- If $|k| < 1$, then $R^n(z) \to 0$.
- If $|k| > 1$, then $R^n(z) \to \infty$.
- If $|k| = 1$ and k is a root of unity, it cycles over a finite set of points.
- If $|k| = 1$ and k is not a root of unity, the subset will be dense in a circle $|z| = |z_0|$.

For a general map with two fixed points ζ_1, ζ_2, we conjugate via $g(z) = \frac{z-\zeta_1}{z-\zeta_2}$ to place the fixed points at 0 and ∞, yielding the same behavioral classification. □

1.3 Iteration of $z \mapsto z^2$

Definition 1.2 (Forward and Backward Invariant)**.** Given a set $E \subseteq \mathbb{C}_\infty$ and a rational function $R : \mathbb{C}_\infty \to \mathbb{C}_\infty$, we say E is:

1. *Forward invariant* if $R(E) \subseteq E$.
2. *Backward invariant* if $R^{-1}(E) \subseteq E$.

Definition 1.3 (Periodic Point)**.** Given a rational function $R : \mathbb{C}_\infty \to \mathbb{C}_\infty$ and $z \in \mathbb{C}_\infty$, we say z is *periodic with period* k if $R^k(z) = z$.

In the Julia set of $z \mapsto z^2$ (which is $J = \{z : |z| = 1\}$), two close points may behave radically differently. And it has this properties:

1. Given I an arc in J, $\exists N \in \mathbb{N}_0$ such that $\forall n \geqslant N$, $R^n(I) = J$.
2. $\exists z_0 \in J$ such that $\{R^n(z_0) \mid n \in \mathbb{N}_0\}$ is dense in J.

3. $\exists z_0 \in J$ such that $\{R^n(z_0) \mid n \in \mathbb{N}_0\}$ is finite.

1.4 Tchebychev Polynomials

Theorem 1.3 (Characterization of Tchebychev polynomials). *Suppose that T is a polynomial of degree k, where $k \geqslant 2$. Then, the interval $[-1, 1]$ is both forward and backward invariant under T if and only if T is T_k or $-T_k$.*

Proof. See [3, pp. 11–12] □

1.5 Iteration of $z \mapsto z^2 - 1$

For polynomials like $P(z) = z^2 - 1$, points outside a certain radius escape to infinity ($z \in F_\infty \implies P^n(z) \to \infty$).

$P(F_0) = F_{-1}$, $P(F_{-1}) = F_0$.

$P(\Omega) \to \ldots, F_0, F_{-1}, F_0, \ldots$, Ω a Fatou set $\neq F_\infty$.

1.6 Iteration of $z \mapsto z^2 + c$

Recall Vieta's formulas. Given a polynomial $P(x) = ax^2 + bx + c$ with roots r_1, r_2:

$$r_1 + r_2 = -\frac{b}{a}, \quad r_1 r_2 = \frac{c}{a}.$$

In general, given a polynomial of degree n, $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$

$$\sum_{i=1}^{n} r_i = -\frac{a_{n-1}}{a_n}, \quad \prod_{i=1}^{n} r_i = (-1)^n \frac{a_0}{a_n},$$

and

$$\sum_{1 \leqslant i < j \leqslant n} r_i r_j = \frac{a_{n-2}}{a_n}, \qquad \sum_{1 \leqslant i < j < k \leqslant n} r_i r_j r_k = \frac{a_{n-3}}{a_n}, \qquad \cdots$$

A second degree polynomial can be transformed into a composition of two affine transformations and z^2:

$$P(z) = z - z^2 = -(z^2 - z) = -\left(z^2 - z + \frac{1}{4} - \frac{1}{4}\right) = -\left(\left(z - \frac{1}{2}\right)^2 - \frac{1}{4}\right).$$

Then defining

$$h(z) = z - \frac{1}{2} \quad , \quad g(z) = z^2 \quad , \quad f(z) = -\left(z - \frac{1}{4}\right) = \frac{1}{4} - z,$$

we have

$$P(z) = f \circ g \circ h(z) = f \circ g \left(z - \frac{1}{2}\right) = f\left(\left(z - \frac{1}{2}\right)^2\right) = -\left(\left(z - \frac{1}{2}\right)^2 - \frac{1}{4}\right) = -(z^2 - z).$$

1.7 Iteration of $z \mapsto z + 1/z$

We consider the dynamics of the rational map defined by:

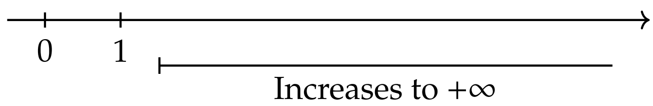
$$R(z) = z + \frac{1}{z}$$

Real Axis:

For real values $z = x \in \mathbb{R}$ with $x > 1$:

$$R(x) = x + \frac{1}{x} > x.$$

The sequence of iterates increases monotonically towards $+\infty$.

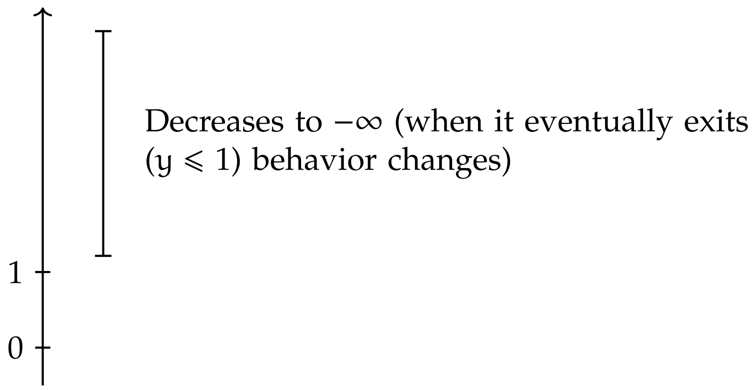


Imaginary Axis:

For purely imaginary values $z = iy \in i\mathbb{R}$ with $y > 1$:

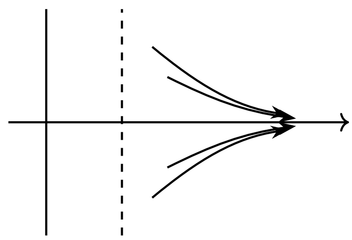
$$R(iy) = iy + \frac{1}{iy} = i\left(y - \frac{1}{y}\right).$$

Since $y > 1$, $y - \frac{1}{y} < y$. Thus, the imaginary part decreases towards 0 as long as $y > 1$. Once it eventually exits this regime ($y \leqslant 1$), the behavior changes completely.



Region $x > 1$:

In the half-plane where $\mathrm{Re}(z) > 1$, the trajectories flow outwards towards the right along curves that flatten asymptotically.



Analyzing the Conjugate Map:

We can understand the behavior at infinity by conjugating via $g(z) = \frac{1}{z}$, which maps the neighborhood of ∞ to a neighborhood of the origin. Let $S = g \circ R \circ g^{-1}$. Since $g^{-1}(z) = \frac{1}{z}$, we have:

$$S(z) = g\left(R\left(\frac{1}{z}\right)\right) = \frac{1}{\frac{1}{z} + z} = \frac{z}{1+z^2},$$

where $g = \frac{1}{z}$. See figure 1.1 for the action of R near ∞.

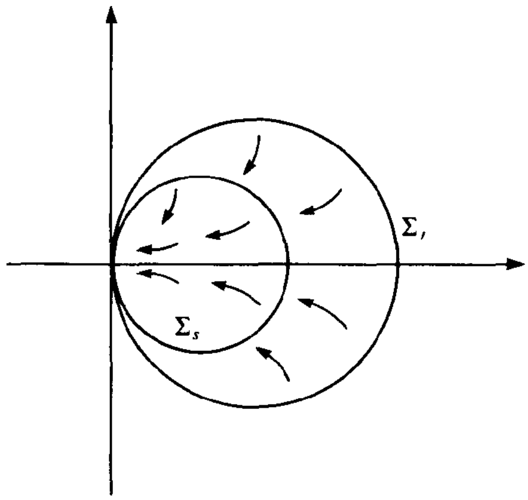


Figure 1.1: Action of $z \mapsto z + 1/z$ near ∞.

1.8 Iteration of $z \mapsto 2z - 1/z$

For $R(z) = 2z - 1/z$, we find the fixed points:

- $\zeta_1 = 1$ (repelling),
- $\zeta_2 = -1$ (repelling),
- $\zeta_3 = \infty$ (attracting).

1.9 Newton's Approximation

To find the roots of a polynomial function f in $\mathbb{R}$, the Newton-Raphson method uses the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Given f in $\mathbb{C}$, we can define the rational function:

$$R(z) = z - \frac{f(z)}{f'(z)}.$$

And the analysis of these kinds of functions (rational functions), falls within the scope of this text.

Note that, in $\mathbb{R}$ that kind of approximation works, but in $\mathbb{C}_\infty \ldots$ the conditions of z_0 to converge to a root are hard and have to do with the

Julia set of R (which for polynomials f with degree 3 or more is usually a complex fractal).

Properties of the Newton method:

1. The roots of f correspond with fixed points of R (proved in the book [3]).

2. Successive iterates of R have a tendency to converge to an attracting fixed point of R.

Thus, if ζ is a zero of f which gives rise to an attracting fixed point of F, then,

$$z_n = F^n(z_0) \xrightarrow[n\to\infty]{} \zeta$$

for at least initial guesses z_0 of ζ sufficiently accurate.

When z_0 is far from ζ the problem of deciding if $z_n \to \zeta$ is much harder.

In the book ([3]) there is analysis of polynomials of degree 2 with distinct roots.

A formulation of a theorem for polynomial with real distinct roots: z_n converges to a root of P for essentially every z_0 (except from a set with area 0)

1.10 General Remarks

For typical rational functions R, the extended complex plane $\mathbb{C}_\infty$ is divided into J (Julia set) and F (Fatou set).

- F preserves proximity.

- J is in every sense chaotic.

- J and F are both forward and backward invariant.

Chapter 2

Rational Maps

We present here some of the elementary properties of rational maps. A point ∞ is adjoined to the complex plane and a rational map of degree d is considered as a d-fold map of the extended complex plane onto itself. The spherical and chordal metrics are introduced, the important notions of conjugacy, valency, fixed points and critical points are discussed, and the Riemann-Hurwitz relation is proved.

2.1 The Extended Complex Plane

Definition 2.1 (Extended Complex Plane)**.** The *extended complex* plane is the union

$$\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\},$$

where ∞ is an "abstract" point outside $\mathbb{C}$.

Now let's define a metric in $\mathbb{C}_\infty$ that handles all points with equal ease and makes ∞ lose any special significance: *the chordal metric*.

Let's start by defining the *Riemann Sphere*. For that, we identify $\mathbb{C}$ with the horizontal plane

$$\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_3 = 0\}$$

in $\mathbb{R}^3$. Let S be the unit sphere in $\mathbb{R}^3$ centered at the origin, and $N = (0, 0, 1)$ be the north pole.

We now project each point in $\mathbb{C}$ to a point in S by taking the intersection between the line that crosses the point $(x_1, x_2, 0)$ and N with S. An illustration of this projection is given in Figure 2.1.

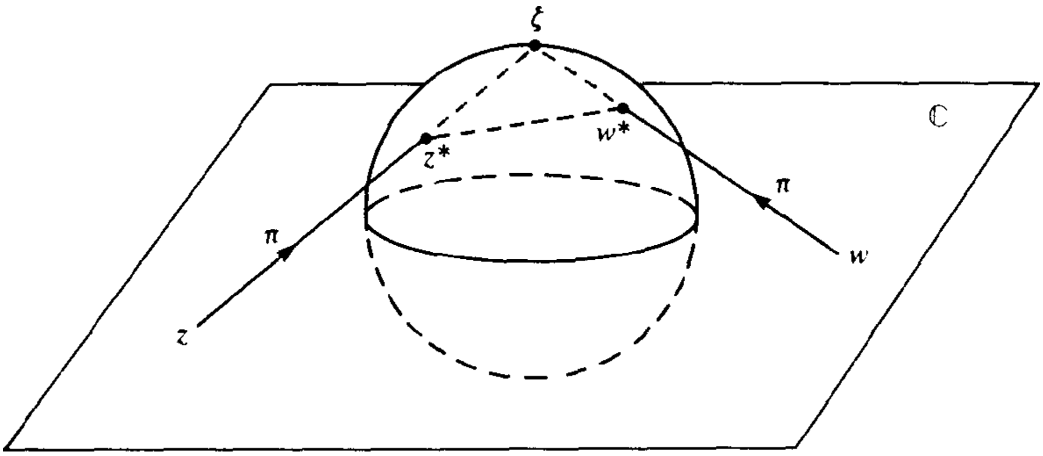


Figure 2.1: Stereographic projection.

We identify this projection as:

$$\pi : \mathbb{C}_\infty \longrightarrow S,$$

where for $z = x + iy \in \mathbb{C}$:

$$\pi(z) = \left(\frac{2x}{1+|z|^2}, \frac{2y}{1+|z|^2}, \frac{|z|^2-1}{1+|z|^2} \right),$$

and $\pi(\infty) = (0, 0, 1)$.

Definition 2.2 (Chordal Metric)**.** Given the projection π from $\mathbb{C}_\infty$ to S, we define the *chordal metric*, $\sigma : \mathbb{C}_\infty \times \mathbb{C}_\infty \longrightarrow \mathbb{R}^+ \cup \{0\}$, as the Euclidean distance from $\pi(z)$ to $\pi(w)$:

$$(z, w) \mapsto \sigma(z, w) = |\pi(z) - \pi(w)|.$$

Explicitly, when $z, w \in \mathbb{C}$:

$$\sigma(z, w) = \frac{2|z-w|}{(1+|z|^2)^{1/2}(1+|w|^2)^{1/2}}.$$

For the point at infinity, we have:

$$\sigma(z, \infty) = \lim_{w\to\infty} \sigma(z, w) = \frac{2}{(1+|z|^2)^{1/2}}.$$

Remark 2.1 ($z \mapsto 1/z$ is an isometry). $\sigma(z, w) = \sigma(z^{-1}, w^{-1})$, i.e., the map $z \mapsto \frac{1}{z}$ is an isometry.

Remark 2.2 (Rotation in $S \mapsto$ isometry in $\mathbb{C}_\infty$). Any rotation $\varphi : S \to S$ of the sphere induces a map $\pi^{-1}\varphi\pi : \mathbb{C}_\infty \to \mathbb{C}_\infty$ which is an isometry. These isometries are Möbius transformations of the form:

$$z \mapsto \frac{az - \bar{c}}{cz + \bar{a}},$$

where $|a|^2 + |c|^2 = 1$.

The chordal metric handles the point at infinity with ease, but it measures distance through the three-dimensional space containing the Riemann sphere. If we restrict our measurements strictly to the two-dimensional curved surface of the sphere, we obtain the spherical metric.

Definition 2.3 (Spherical Metric)**.** Given the stereographic projection $\pi : \mathbb{C}_\infty \to S$, the *spherical distance* $\sigma_0(z, w)$ between two points in $\mathbb{C}_\infty$ is defined as the Euclidean length of the shortest path (an arc of a great circle) connecting their projections $\pi(z)$ and $\pi(w)$ on the surface of S.

If the chord joining the projected points subtends an angle θ at the origin of the sphere, then by definition the spherical distance is exactly $\sigma_0(z, w) = \theta$. By using the trigonometric connection to the chordal metric, we can derive explicit formulas directly in terms of the complex coordinates for $z, w \in \mathbb{C}$:

$$\sigma(z, w) = 2\sin\left(\frac{\theta}{2}\right) = 2\sin\left(\frac{\sigma_0(z, w)}{2}\right).$$

Thus,

$$\begin{aligned} \sigma_0(z, w) &= 2\arcsin\left(\frac{\sigma(z, w)}{2}\right) \\ &= 2\arcsin\left(\frac{|z - w|}{(1 + |z|^2)^{1/2}(1 + |w|^2)^{1/2}}\right). \end{aligned}$$

For the point at infinity, taking the limit as $w \to \infty$ yields:

$$\sigma_0(z, \infty) = 2\arcsin\left(\frac{1}{(1 + |z|^2)^{1/2}}\right).$$

Remark 2.3 ($\sigma \sim \sigma_0$). The chordal and spherical metrics are topologically equivalent ($\sigma(z, w) = 2\sin(\theta/2)$). Furthermore, they satisfy the following inequalities:

$$\frac{2}{\pi}\sigma_0(z, w) \leqslant \sigma(z, w) \leqslant \sigma_0(z, w)$$

Proof. The inequalities follow directly from applying the elementary real-variable inequality $\frac{2\theta}{\pi} \leqslant \sin\theta \leqslant \theta$ on the interval $0 \leqslant \theta \leqslant \frac{\pi}{2}$ to the angle $\theta = \frac{\sigma_0(z,w)}{2}$. □

Because these metrics differ by at most a bounded scale factor of $\frac{2}{\pi}$, we can use σ and σ_0 interchangeably throughout our study of rational dynamics whenever global constants are allowed to change.

Remark 2.4 (Spherical length). Given a continuously differentiable curve $\gamma : I \longrightarrow \mathbb{C}_\infty$, its *spherical length* is given by:

$$\int_\gamma \frac{2|dz|}{1 + |z|^2}.$$

2.2 Rational Maps

Definition 2.4 (Rational Map)**.** A *rational map* is a function of the form

$$R(z) = \frac{P(z)}{Q(z)},$$

where $P(z)$ and $Q(z)$ are polynomials, not both being the zero polynomial.

- If $P = 0 \Rightarrow R = 0$.
- If $Q = 0 \Rightarrow R = \infty$.
- If $P(z_0) \neq 0$ and $Q(z_0) = 0$ for some $z_0 \in \mathbb{C}$, then $R(z_0) = \infty$.
- We define $R(\infty) = \lim_{z\to\infty} R(z)$.

- If $P(z_0) = 0$ and $Q(z_0) = 0$, strictly speaking $R(z_0)$ is not defined, but in this case we may cancel the common factors:

$$P(z) = (z - z_0)^m P_1(z)$$
$$Q(z) = (z - z_0)^n Q_1(z)$$

where $P_1(z_0) \neq 0$ and $Q_1(z_0) \neq 0$. Then:

$$R(z) = \frac{P_1(z)}{Q_1(z)}(z - z_0)^{m-n}.$$

From now on, we will consider that this has been done, i.e., P and Q are coprime. Thus, given R, the polynomials P and Q are uniquely determined up to scalar multiplication. This allows us to define the degree of a rational map.

Definition 2.5 (Degree of a Rational Map, $\deg(R)$)**.** Given a rational map $R = P/Q$ where P and Q are coprime polynomials, the *degree of* R is defined as:

$$\deg(R) = \max\{\deg(P), \deg(Q)\}$$

where $\deg(\cdot)$ is the usual degree of a polynomial. We will consider that if $R = \alpha \in \mathbb{C}_\infty$, then $\deg(R) = 0$.

Proposition 2.5 ($\deg(RS) = \deg(R)\deg(S)$)**.** *Given two rational function* R*,* S*, the degree of* RS *is* $\deg(R)$ *times* $\deg(S)$*. That is,*

$$\deg(RS) = \deg(R)\deg(S).$$

Remember that by writing RS *we mean composition (*$R \circ S$*) not multiplication. However,* $\deg(R)\deg(S)$ *is a multiplication.*

Proof. See [3, p. 32]. ☐

Corollary 2.5.1 ($\deg(R^n) = [\deg(R)]^n$)**.** *Given a rational function* R*, the degree of* R^n *is the* n*-th power of* $\deg(R)$*. That is,*

$$\deg(R^n) = [\deg(R)]^n.$$

Now, let's remember the definitions of *holomorphic*, *meromorphic* and *analytic* functions in $\mathbb{C}$ and see how they translate to $\mathbb{C}_\infty$ with the chordal metric.

Definition 2.6 (Holomorphic)**.** Given $D \subseteq \mathbb{C}$, a function $f : D \to \mathbb{C}$ is *holomorphic* in D if the derivative of f exists at each point in D.

Definition 2.7 (Meromorphic)**.** Given $D \subseteq \mathbb{C}$, a function $f : D \to \mathbb{C}_\infty$ is *meromorphic* in D if each point in D has a neighborhood on which either f or $1/f$ is holomorphic.

Definition 2.8 (Pole of f)**.** Given a meromorphic function f in $D \subseteq \mathbb{C}$, the *poles of* f are the points $w \in D$ where $f(w) = \infty$ ($\lim_{z\to w} f(z) = \infty$).

Remark 2.6 (f meromorphic $\implies 1/f$ holomorphic at f's poles). Given a meromorphic function f in $D \subseteq \mathbb{C}$ with a pole at w, then $1/f$ is holomorphic near w and $\frac{1}{f(w)} = 0$.

Definition 2.9 (f is defined near ∞)**.** A function f is said to be *defined near* ∞ (or *in a neighborhood of* ∞) if it is defined on some set $\{z \in \mathbb{C} \mid |z| > r\} \cup \{\infty\}$.

Definition 2.10 (Holomorphic or Meromorphic at ∞)**.** Given a function f defined near ∞, f is *holomorphic* (or *meromorphic*) *at* ∞ if $f(1/z)$ is holomorphic (or meromorphic) near 0.

Definition 2.11 (Analytic in $D \subseteq \mathbb{C}_\infty$)**.** Given a map $f : D_1 \to D_2$ between any two subdomains of $\mathbb{C}_\infty$, we say f is *analytic* if it is holomorphic or meromorphic at each point in D_1.

Remark 2.7 (f meromorphic $\implies$ f continuous at its poles). Given a meromorphic function f in $D \subseteq \mathbb{C}$ with a pole at w, then (using the chordal metric and its topology), f is continuous at the pole. This is because $z \mapsto 1/z$ is a σ-isometry:

$$\sigma(f(z), f(w)) = \sigma\left(\frac{1}{f(z)}, \frac{1}{f(w)}\right) = \sigma\left(\frac{1}{f(z)}, 0\right) = \frac{2|1/f(z)|}{(1+|1/f(z)|^2)^{1/2}} \xrightarrow{z\to w} 0$$

Summing up, we have that the concepts of *holomorphic*, *meromorphic* and *analytic* functions translate well from $\mathbb{C}$ with the Euclidean metric, to $\mathbb{C}_\infty$ with the chordal or spheric metric. Hence, making ∞ lose any special significance it may have had in $\mathbb{C}$.

Proposition 2.8 (Polynomials are analytic in $\mathbb{C}_\infty$)**.** *Given a polynomial* $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ *where* $n > 0$ *and* $a_n \neq 0$*. Then,* P *is analytic in* $\mathbb{C}_\infty$*.*

Proof. This proof serves to illustrate what holomorphic and meromorphic mean. We have to show that P is holomorphic or meromorphic at each point in $\mathbb{C}_\infty$. We have that P is analytic in $\mathbb{C}$ because it's a polynomial, thus, we only have to show that P is holomorphic or meromorphic at ∞. We'll see that it's meromorphic at ∞, i.e., $P(1/z)$ is meromorphic at 0, i.e., $1/P(1/z)$ is holomorphic at 0. And we can see that

$$\frac{1}{P(1/z)} = \frac{z^n}{a_0 z^n + \cdots + a_n}$$

is differentiable at 0, therefore, P is meromorphic at ∞. □

In this proof we have also seen that all polynomials have a pole in ∞, that is, $P(\infty) = \infty$. We will see that this means that ∞ is a fixed point for all polynomials.

Theorem 2.9 (Analytic in $\mathbb{C}_\infty \iff$ rational map)**.** *The rational maps are precisely the analytic maps of* $\mathbb{C}_\infty$ *into itself.*

Theorem 2.10 (Rational maps are d-fold maps in $\mathbb{C}_\infty$)**.** *Given* R *a rational function of positive degree* d*. Then* R *is a* d*-fold map of* $\mathbb{C}_\infty$ onto *itself: that is, for any* w *in* $\mathbb{C}_\infty$*, the equation* $R(z) = w$ *has* precisely d *solutions in* z *(counting multiplicities).*

Proof. See [3, pp. 31–32]. □

Remark 2.11 (Zeros and poles of rational maps). Given $R = P/Q$ a rational function of positive degree d, where P and Q are coprime polynomials with $\deg(P) = n$ and $\deg(Q) = m$. Then, the zeros of R are the zeros of P, and the poles of R are the zeros of Q. Moreover, the number of zeros and poles of R at ∞ (counting multiplicities) are given by the following cases:

1. if $n = m$, then all the zeros and poles of R are in $\mathbb{C}$;
2. if $n > m$, then the number of poles at ∞ is $n - m$;
3. and if $n < m$, then the number of zeros at ∞ is $m - n$.

2.3 The Lipschitz Condition

Theorem 2.12 (Rational maps satisfy a Lipschitz condition)**.** *A rational map* R *satisfies some* Lipschitz condition

$$\sigma_0(R(z), R(w)) \leqslant M\sigma_0(z, w)$$

on the complex sphere.

Proof. See [3, p. 33]. □

Theorem 2.13 (Möbius transformations explicit Lipschitz condition)**.** *The Möbius map*

$$g(z) = \frac{az + b}{cz + d}, \quad ad - bc = 1,$$

satisfies the Lipschitz condition

$$\sigma_0(g(z), g(w)) \leqslant ||g||^2 \sigma_0(z, w).$$

Proof. See [3, pp. 33–34]. □

Theorem 2.14 (Möbius transformations *family* explicit uniform Lipschitz condition)**.** *Let m be any positive number. Then the Möbius transformations g which satisfy*

$$\begin{aligned}\sigma(g(0),\ g(1)) &\geqslant m,\\ \sigma(g(1),\ g(\infty)) &\geqslant m,\\ \sigma(g(\infty), g(0)) &\geqslant m\end{aligned}$$

also satisfy the uniform Lipschitz condition

$$\sigma(g(z), g(w)) \leqslant \frac{\pi}{m^3}\sigma(z, w).$$

Proof. See [3, pp. 34–35]. □

Definition 2.12 (Complementary components of a closed curve γ)**.** Let γ be a *closed curve* on the sphere. The complement of γ is a union of mutually disjoint domains, which we call the *complementary components* of γ.

Definition 2.13 (Exterior and interior of a closed curve γ contained in a open hemisphere)**.** Let γ be a *closed curve* that lies in some *open hemisphere*. Then exactly one of the complementary components contains a hemisphere; we denote this by $E(\gamma)$ and call it the *exterior* of γ. The other complementary components of γ are said to be the *interior components* of γ.

Theorem 2.15 (Interior components mapping theorem for rational maps)**.** *Let R be a rational function. Then there is a positive number δ such that if γ is any closed curve of σ_0-diameter less than δ, then the image $R(\Omega)$ of an interior component Ω of γ does not meet the exterior $E(R(\gamma))$ of $R(\gamma)$.*

Proof. See [3, pp. 35–36]. □

This result almost says that if the curve γ is sufficiently small, then the interior components of γ map to the interior components of the image curve $R(\gamma)$: however, the images of the interior components may also contain points of $R(\gamma)$, so it is necessary to express this a little more carefully in terms of the exterior of the image curve (Figure 2.2).

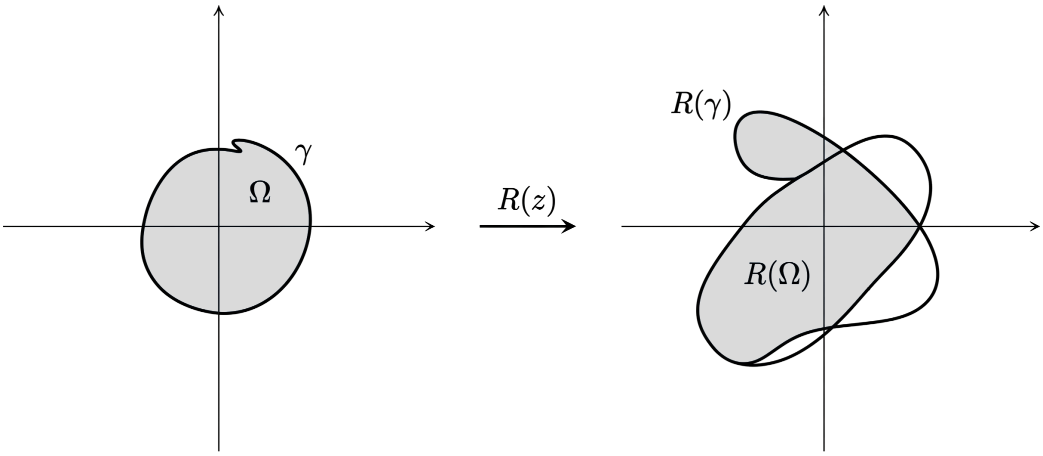


Figure 2.2: If the curve γ is sufficiently small ($\mathrm{diam}(\gamma) < \delta$), the image of its interior component Ω under a rational function R is entirely contained within the bounded regions defined by the image curve $R(\gamma)$, never entering the exterior $E(R(\gamma))$.

2.4 Conjugacy

Definition 2.14 (Conjugate rational maps)**.** Given two rational maps R and S, we say R and S are *conjugate* if there is some Möbius map g with

$$S = gRg^{-1}.$$

Remark 2.16 (Conjugacy is an equivalence relation). Conjugacy is an equivalence relation, and the equivalence classes are *conjugacy classes* of rational maps.

Remark 2.17 (Invariant properties of conjugacy). Given two conjugate rational maps R and S, where $S = gRg^{-1}$. Then, the following are true:

1. $\deg(R) = \deg(S)$;

2. $S^n = gR^n g^{-1}$ (this means that we can transfer a problem concerning R to a (possibly simpler) problem concerning a conjugate S);

3. and S fixes $g(z)$ if and only if R fixes z (conjugacy respects fixed points).

All of these observations are elementary; nevertheless, they are important because, in general, we shall not bother to distinguish between conjugate rational functions.

As a simple illustration of the use of conjugacy, let us characterize the polynomials within the class of rational maps.

Remark 2.18 (Characterization of polynomials). A rational map R is a polynomial if and only if R has a pole at ∞ and no poles in $\mathbb{C}$; more succinctly, if and only if $R^{-1}\{\infty\} = \{\infty\}$.

Theorem 2.19 (Characterization of the conjugacy class of the polynomials)**.** *A non-constant rational map R is conjugate to a polynomial if and only if there is some w in $\mathbb{C}_\infty$ with $R^{-1}\{w\} = \{w\}$.*

2.5 Valency

Definition 2.15 (Valency of f at $z_0 \in \mathbb{C}$)**.** Given any function f that is non-constant and holomorphic near the point z_0 in $\mathbb{C}$. Then f has a Taylor expansion at z_0, say

$$f(z) = f(z_0) + a_k(z - z_0)^k + a_{k+1}(z - z_0)^{k+1} + \dots$$

where $a_k \neq 0$ and the positive integer k is uniquely determined by the condition that the limit

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{(z - z_0)^k}$$

exists, is finite, and is non-zero.

We denote this integer k by $\nu_f(z_0)$ and call it the *valency*, or *order*, of f at z_0: of course, it is the number of solutions of $f(z) = f(z_0)$ at z_0.

Proposition 2.20 (Chain Rule for Valency)**.** *The valency function ν satisfies the important* Chain Rule*:*

$$\nu_{fg}(z_0) = \nu_f(g(z_0))\nu_g(z_0),$$

where f and g are non-constant holomorphic functions near $g(z_0)$ and z_0 respectively; and z_0, $g(z_0)$, and $fg(z_0)$ are all in $\mathbb{C}$.

Proof. See [3, pp. 37–38]. □

Remark 2.21 (Injectivity in terms of valency). Given any function f that is non-constant and holomorphic near the point z_0 in $\mathbb{C}$. Observe that f is injective in some neighborhood of z_0 if and only if $\nu_f(z_0) = 1$.

Remark 2.22 (Composing with injective functions preserves valency). Because of the previous remark (2.21), the Chain Rule shows that the valency is preserved under a pre-application, and a post-application, of an injective function.

Explicitly, if fg is defined near z with g injective near z, and if hf is defined near z with h injective near $f(z)$, then:

$$\nu_{fg}(z) = \nu_f(g(z)),$$
$$\nu_{hf}(z) = \nu_f(z).$$

The properties of valency under injective functions enable us to extend our definition of $\nu_f(z_0)$ to the case where $z_0 = \infty$, or $f(z_0) = \infty$ (or both).

Definition 2.16 (Valency at ∞)**.** Given any function f that is non-constant and holomorphic near the point ∞ in $\mathbb{C}_\infty$ or z_0 in $\mathbb{C}_\infty$ such that $f(z_0) = \infty$ (or both). Then, we can select any Möbius maps g and h which map z_0 and $f(z_0)$ respectively to points in $\mathbb{C}$ and then define $\nu_f(z_0)$ by

$$\nu_f(z_0) = \nu_F(g(z_0)),$$

where $F = hfg^{-1}$.

The essential observation is that if we repeat this procedure using different maps g and h, then the resulting answer will be the same; thus $\nu_f(z_0)$ is defined independently of the choice of g and h.

Definition 2.17 (Univalent map)**.** An analytic map $f : D \to \mathbb{C}_\infty$ is said to be *univalent* in D if it is injective there.

Remark 2.23 (f univalent $\implies$ $v_f(z) = 1 \forall z \in D$). If f is univalent in D, then $v_f(z) = 1$ for every z in D.

The converse *need not hold*, being univalent is a global property (the whole domain), but valency is a local property (a neighborhood of z_0).

Example 2.1. The map $z \mapsto z^2$ of $\mathbb{C} - \{0\}$ onto itself has valency 1 everywhere, but it is not injective in $\mathbb{C} - \{0\}$.

Remark 2.24 (Rational maps are d-fold maps via valency). We've seen that rational maps are d-fold maps (Theorem 2.10), now we can write this in terms of valency. Given a non-constant rational map R of degree d, then for any z_0 in $R^{-1}\{w\}$, $v_R(z_0)$ is the number of solutions of $R(z) = w$ at z_0, and so for each w in the complex sphere,

$$\sum_{z \in R^{-1}\{w\}} v_R(z) = \deg(R).$$

2.6 The Inverse Function Theorem

Now that we have established the local behavior of analytic maps via valency, we can formalize when these maps can be locally inverted. In the classical complex plane, this is governed by the derivative.

Theorem 2.25 (Inverse Function Theorem in $\mathbb{C}$)**.** *Let $f : U \to \mathbb{C}$ be a holomorphic function on an open set $U \subseteq \mathbb{C}$, and let $z_0 \in U$. If $f'(z_0) \neq 0$, then there exist open neighborhoods $V \subseteq U$ of z_0 and W of $f(z_0)$ such that the restriction $f : V \to W$ is a bijection.*

Furthermore, the local inverse function $f^{-1} : W \to V$ is holomorphic, and its derivative satisfies:

$$(f^{-1})'(f(z_0)) = \frac{1}{f'(z_0)}.$$

Let us now bridge this classical theorem with our structural definition of valency at a point.

Remark 2.26 (Local injectivity in terms of derivative and valency in $\mathbb{C}$). By examining the Taylor expansion of a non-constant holomorphic function f near $z_0 \in \mathbb{C}$:

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \cdots$$

we see that $f'(z_0) \neq 0$ if and only if the first non-zero coefficient appears at index $k = 1$. By Definition 2.15, this means $\nu_f(z_0) = 1$.

Thus, the Inverse Function Theorem in $\mathbb{C}$ (Theorem 2.25) states that f possesses a local holomorphic inverse near $z_0 \in \mathbb{C}$ if and only if $\nu_f(z_0) = 1$.

Just as we did with valency, we can exploit the fact that composing with injective functions preserves local bijectivity to extend this theorem to cases where $z_0 = \infty$, $f(z_0) = \infty$, or both.

Theorem 2.27 (Inverse Function Theorem in $\mathbb{C}_\infty$)**.** *Let $f : D_1 \to D_2$ be a non-constant analytic map between two subdomains of $\mathbb{C}_\infty$, and let $z_0 \in D_1$. Then, we can select any two Möbius transformations g and h which map z_0 and $f(z_0)$ respectively to points in $\mathbb{C}$. Let $F = hfg^{-1}$ be the localized map defined near $g(z_0) \in \mathbb{C}$. If $F'(g(z_0)) \neq 0$, then f is a local bijection from a σ-neighborhood of z_0 onto a σ-neighborhood of $f(z_0)$. Consequently, it possesses a local analytic inverse f^{-1} near $f(z_0)$.*

The essential observation is that if we repeat this procedure using different maps g and h, then the resulting answer will be the same; thus the theorem can be generalized to $\mathbb{C}_\infty$ from $\mathbb{C}$ independently of the choice of g and h.

Remark 2.28 (Local injectivity in terms of localized derivative and valency in $\mathbb{C}_\infty$). Let f, g, h, and F be defined as in Theorem 2.27. By expanding the non-constant holomorphic map F into a Taylor series near the point $w_0 = g(z_0) \in \mathbb{C}$:

$$F(w) = F(w_0) + F'(w_0)(w - w_0) + \frac{F''(w_0)}{2!}(w - w_0)^2 + \cdots$$

we observe that $F'(g(z_0)) \neq 0$ if and only if the first non-zero coefficient after the constant term appears at index $k = 1$.

By your definition of extended valency, this means $\nu_F(g(z_0)) = \nu_f(z_0) =$

1. Thus, the Inverse Function Theorem in $\mathbb{C}_\infty$ (Theorem 2.27) establishes a universal rule across the entire Riemann sphere: an analytic map f admits a local analytic inverse near a point $z_0 \in \mathbb{C}_\infty$ if and only if $\nu_f(z_0) = 1$.

2.7 Fixed Points

Definition 2.18 (Fixed point of f)**.** Given any function $f : D \to \mathbb{C}_\infty$ defined on some domain $D \subseteq \mathbb{C}_\infty$. A point z_0 in D is a *fixed point* of f if $f(z_0) = z_0$.

Proposition 2.29 (Fixed points of rational maps)**.** *Given by two coprime polynomials* P *and* Q *and the rational map* $R = P/Q$. *Then,*

1. R *fixes* ∞ *if and only if* $\deg(P) > \deg(Q)$*;*
2. *and* R *fixes* $\zeta \in \mathbb{C}$ *if and only if* $P(\zeta) = \zeta Q(\zeta)$*; that is, the* fixed points of R in $\mathbb{C}$ *are the solutions of*

$$P(z) - zQ(z) = 0.$$

Proof. See [3, p. 39]. □

We count fixed points the same way we count zeros, that is, counting multiplicities.

Definition 2.19 (Number of fixed points at $z_0 \in \mathbb{C}$)**.** Given any function $f : D \to \mathbb{C}_\infty$ defined on some domain $D \subseteq \mathbb{C}_\infty$. The number of fixed points of f at a point z_0 in $\mathbb{C}$ is given by the valency of the function $f(z) - z$ at z_0, i.e., $\nu_{f(z)-z}(z_0)$.

Example 2.2. The map $f(z) = z + z^3$ has three fixed points at 0. This is because the fixed points of f are the zeros of $f(z) - z$, and

$$f(z) - z = z^3,$$

which has valency 3 at 0 ($z^3 = f(0) - 0 = 0$ has three solutions). This is not to be confused with the valency of f at 0, which is 1. In fact, $f(z) = f(0)$ has only one solution at 0 ($f(z) = z(1 + z^2)$).

To count the number of fixed points at ∞, we need the following lemma.

Lemma 2.30 ($\zeta \in \mathbb{C}$ fixed by $f \implies \zeta$ fixed by conjugate in $\mathbb{C}$)**.** *Let ζ in $\mathbb{C}$ be a fixed point of an analytic map f, and let φ be any map that is analytic, injective and finite in some neighborhood of ζ. Then, $\varphi f \varphi^{-1}$ has the same number of fixed points at $\varphi(\zeta)$ as f has at ζ.*

Proof. See [3, pp. 39–40]. □

Definition 2.20 (Number of fixed points at ∞)**.** Given any function $f : D \to \mathbb{C}_\infty$ defined on some domain $D \subseteq \mathbb{C}_\infty$, $\infty \in D$. Then we can select an analytic map φ, that is injective and finite in some neighborhood of ∞, and then define the number of fixed points of f at ∞ by the number of fixed points of $\varphi f \varphi^{-1}$ at $\varphi(\infty)$.

The essential observation is that if we repeat this procedure using different maps φ, then the resulting answer will be the same; thus, the number of fixed points of f at ∞ is defined independently of the choice of φ.

Moreover, as a direct consequence of the previous lemma, we have the following theorem.

Theorem 2.31 (Number of fixed points is invariant under conjugacy)**.** *Given two conjugate rational maps R and S, where $S = gRg^{-1}$, g is a Möbius transformation. Then, the number of fixed points of R at ζ is the same as the number of fixed points of S at $g(\zeta)$.*

Theorem 2.32 (Number of fixed points of a rational map)**.** *Given a rational map R of degree $d \geqslant 1$. Then, the number of fixed points of R in $\mathbb{C}_\infty$ is exactly $d + 1$ (counting multiplicities).*

Proof. See [3, p. 40]. □

Definition 2.21 (Multiplier of a fixed point in $\mathbb{C}$)**.** Given a rational map R and a fixed point ζ in $\mathbb{C}$ of R. We define *multiplier* of R at ζ, denoted by $m(R, \zeta)$, as the complex number given by the derivative:

$$m(R, \zeta) = R'(\zeta).$$

Definition 2.22 (Multiplier of the fixed point ∞)**.** Given a rational map R with a fixed point at ∞. Then, we can select a Möbius map g that maps ∞ to some point in $\mathbb{C}$, and then define the multiplier of R at ∞ by the multiplier of gRg^{-1} at $g(\infty)$:

$$m(R, \infty) = m(gRg^{-1}, g(\infty)).$$

Note that, by the preceding remarks, this definition is independent of the choice of g.

Example 2.3 (Multiplier at ∞ for a rational map)**.** Let R be a rational map of the form

$$R(z) = \frac{a_0 + a_1 z + \cdots + a_n z^n}{b_0 + b_1 z + \cdots + b_m z^m},$$

where $a_n b_m \neq 0$ and $n > m$, so R fixes ∞. By definition, $m(R, \infty)$ is the derivative of the conjugate map $S(z) = 1/R(1/z)$ at the origin. A direct computation shows:

$$S'(0) = \begin{cases} b_m/a_n & \text{if } n = m + 1, \\ 0 & \text{if } n > m + 1. \end{cases}$$

On the other hand, computing the derivative $R'(z)$ and taking the limit gives:

$$R'(\infty) = \lim_{z \to \infty} R'(z) = \begin{cases} a_n/b_m & \text{if } n = m + 1, \\ \infty & \text{if } n > m + 1. \end{cases}$$

Remark 2.33 (Formula for the multiplier of a rational map at a fixed point in $\mathbb{C}_\infty$). These results show that the multiplier $m(R, \zeta)$ of a rational map

R at fixed point ζ is given by

$$m(R,\zeta) = \begin{cases} R'(\zeta) & \text{if } \zeta \neq \infty; \\ \frac{1}{R'(\infty)} & \text{if } \zeta = \infty. \end{cases}$$

Example 2.4. For example, if $R(z) = 3z$, then ∞ is an attracting fixed point of R (this is geometrically clear) and in this case, $m(R,\infty) = 1/3$.

We conclude this section by analyzing the behavior of the number of fixed points under iteration near a fixed point of f.

Theorem 2.34 (Iterates near the fixed point 0)**.** *Given an analytic function* f *that in a neighborhood of the origin has the expansion*

$$f(z) = az + b_1 z^{r+1} + \cdots,$$

where $a \neq 0$, $b_1 \neq 0$, *and* $r \geqslant 1$. *Then its* n*-th iterate, in that neighborhood, has the form*

$$f^n(z) = a^n z + b_n z^{r+1} + \cdots,$$

where

$$b_n = a^{n-1} b_1 \left(1 + a^r + a^{2r} + \cdots + a^{(n-1)r}\right).$$

Proof. See [3, p. 42]. □

From this theorem, we immediately obtain several important corollaries regarding the behavior of coefficients and the multiplicity of fixed points under iteration.

Corollary 2.34.1 ($a = 1 \implies b_n = nb_1$)**.** *If* $a = 1$, *then* $b_n = nb_1$.

Proof. This is a direct consequence of the formula for b_n:

$$b_n = a^{n-1} b_1 \left(1 + a^r + a^{2r} + \cdots + a^{(n-1)r}\right) = b_1(1 + 1 + \cdots + 1) = nb_1.$$

□

Corollary 2.34.2 ($b_n = 0 \iff a^r \neq 1$ and $a^{nr} = 1$)**.** $b_n = 0$ *if and only if* $a^r \neq 1$ *and* $a^{nr} = 1$.

Proof. The terms inside the parentheses in the formula for b_n form a geometric series. Thus, we can write:

$$b_n = a^{n-1} b_1 \frac{1 - a^{nr}}{1 - a^r}.$$

For b_n to be zero, the numerator of that fraction must be zero, but the denominator must not be zero (to avoid a $0/0$ situation, which we already showed in the previous corollary (Corollary 2.34.1) that it evaluates to n).

$$\begin{aligned} 1 - a^{nr} &= 0 \implies a^{nr} &= 1, \\ 1 - a^r &\neq 0 \implies a^r &\neq 1. \end{aligned}$$

Thus, $b_n = 0$ if and only if $a^r \neq 1$ and $a^{nr} = 1$. □

Remark 2.35 (f^n has a single fixed point at the origin $\implies$ f has a single fixed point at the origin). If $a^n \neq 1$, then $a \neq 1$. Thus, if f^n has a single fixed point at the origin, then so does f.

Proof. If $a = 1$, then $a^n = 1$ for all n. Thus, if $a^n \neq 1$, then $a \neq 1$.

If $a \neq 1$, the function $f(z) - z$ looks like $(a-1)z + b_1 z^{r+1} + \cdots = z((a-1) + b_1 z^r + \cdots)$, where $a - 1 \neq 0$. Thus, $f(z)$ has a single fixed point at the origin.

If f^n has a single fixed point at the origin, then we can only factor out one z from $f^n(z) - z$, giving us $f^n(z) - z = (a^n - 1)z + b_n z^{r+1} + \cdots = z((a^n - 1) + b_n z^r + \cdots)$, where $a^n - 1 \neq 0$.

Combining these three results, we have that if f^n has a single fixed point at the origin, then $a^n \neq 1$, which implies $a \neq 1$, thus f has a single fixed point at the origin. □

Corollary 2.35.1 (Number of fixed points at 0 of the iterates)**.** *The iterate* f^n *has at least as many fixed points at the origin as* f *has. Furthermore, if* f^n *has strictly more fixed points at the origin than* f*, then* $a \neq 1$ *but* $a^n = 1$.

Proof. Let's begin by showing that f^n has at least as many fixed points at the origin as f has. We have two cases:

- If $a \neq 1$: $f(z) - z$ has a single fixed point at the origin. Since $f^n(0) = 0$, 0 must be a fixed point of f^n of multiplicity at least 1.

- If $a = 1$: then $f(z) - z = b_1 z^{r+1} + \cdots$, so 0 is a fixed point of f of multiplicity $r + 1$ ($b_1 \neq 0$). On the other hand, by Corollary 2.34.1, $f^n(z) - z = nb_1 z^{r+1} + \cdots$, so 0 is a fixed point of f^n of multiplicity $r + 1$ (the same as for f). Thus, if $a = 1$, then f^n has as many fixed points at the origin as f has.

Now, let's show that if f^n has strictly more fixed points at the origin than f, then $a \neq 1$ but $a^n = 1$. From our analysis above, if $a = 1$, the multiplicities are exactly equal, so for f^n to have strictly more fixed points at the origin than f, we cannot have $a = 1$. Thus $a \neq 1$.

If $a \neq 1$, then f has a single fixed point at the origin. For f^n to have strictly more fixed points at the origin than f, the term $(a^n - 1)$ from $f^n(z) - z = (a^n - 1)z + b_n z^{r+1} + \cdots = z((a^n - 1) + b_n z^r + \cdots)$ must vanish so that a higher power of z takes over. Thus, we must have $a^n = 1$. □

2.8 Critical Points

Definition 2.23 (Critical point of a rational map)**.** Given a rational map R. A point z_0 in $\mathbb{C}_\infty$ is a *critical point* of R if R fails to be injective in any neighborhood of z_0, and if R is not constant. These points are precisely the points at which $v_R(z_0) > 1$.

Definition 2.24 (Critical value of a rational map)**.** Given a rational map R. A point w_0 in $\mathbb{C}_\infty$ is a *critical value* of R if it is the image of some critical point; that is, if $w_0 = R(z_0)$ for some critical point z_0 of R.

Remark 2.36. Given a rational function R of degree $d \geqslant 1$, and w a non-critical value of R. Then, $R^{-1}\{w\}$ consists of precisely d distinct points, say $z_1, z_2, \ldots, z_d$. As none of the z_j are critical points, there are neighborhoods of N of w, and $N_1, \ldots, N_d$ of $z_1, \ldots, z_d$ respectively, with R acting as a bijection from each N_j onto N. See Figure 2.3. It follows that for

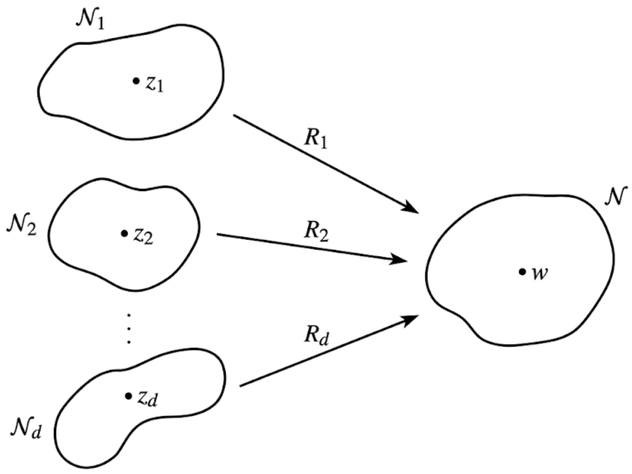


Figure 2.3: Bijective restrictions of R for a non-critical value w.

each j, the restriction R_j of R to N_j ($R : N_j \to N$) has an inverse

$$R_j^{-1} : N \to N_j.$$

Definition 2.25 (Branch of R^{-1} at a non-critical value w)**.** Given a rational function R of degree $d \geqslant 1$, and w a non-critical value of R. Then, we call each of the inverse maps $R_j^{-1} : N \to N_j$ a *branch* of R^{-1} at w.

By applying the Inverse Function Theorem (Theorem 2.25), we know that R is injective on some neighborhood of any point in $\mathbb{C}$ at which R' has neither a zero nor a pole: thus for all but a finite set of z, $\nu_R(z) = 1$ (recall that locally injective is the same as having valency 1 (Remark 2.21)) and, consequently,

$$\sum_{z \in \mathbb{C}_\infty} [\nu_R(z) - 1] < \infty.$$

This sum gives us a measure of the number of multiple roots of R (and the difficulties in defining R^{-1}), and its actual value is given by the *Riemann-Hurwitz relation*.

Theorem 2.37 (Riemann-Hurwitz relation)**.** *Given a non-constant rational map* R*. Then,*

$$\sum_{z \in \mathbb{C}_\infty} [\nu_R(z) - 1] = 2d - 2.$$

Proof. See [3, p. 44]. □

The general term in the sum is positive only when z is a critical point of R so, as the valency is an integer, this provides us with an estimate of the number of critical points of R. For a non-constant polynomial P, we have $\nu_P(\infty) = \deg(P)$, so, in the same way, we obtain an estimate of the number of critical points of P in $\mathbb{C}$. These two estimates are given in the following corollary.

Corollary 2.37.1 (Bound on amount of critical points of a rational)**.** *A rational map of positive degree* d *has at most* $2d-2$ *critical points in* $\mathbb{C}_\infty$*. A polynomial of positive degree* d *has at most* $d-1$ *critical points in* $\mathbb{C}$*.*

Proof. Given a non-constant rational map R of degree d, for a point to be critical we must have $\nu_R(z) > 1$, so $\nu_R(z) - 1 \geqslant 1$. Thus, the number of critical points of R in $\mathbb{C}_\infty$ is at most $2d-2$.

In the case of polynomials, we have $\nu_P(\infty) = \deg(P) = d$, so the contribution of ∞ to the sum in the Riemann-Hurwitz relation is $d-1$. Thus, the number of critical points of P in $\mathbb{C}$ is at most $(2d-2)-(d-1) = d-1$. □

We end this section with a simple result which locates the critical values of the iterate R^n in terms of the critical points of R.

Theorem 2.38 (Critical values of R^n)**.** *Given a rational map* R *and its set of critical points* C*. Then, the set of critical values of* R^n *is*

$$R(C) \cup \cdots \cup R^n(C).$$

Proof. See [3, p. 45]. □

2.9 A Topology on the Rational Functions

To study the global dynamics and stability of families of rational maps, we must equip the set of rational functions with a robust topological structure.

Definition 2.26 (Spaces $\mathcal{C}$ and $\mathcal{R}$)**.** Let $\mathcal{C}$ denote the class of all continuous maps from the extended complex plane into itself:

$$\mathcal{C} = \{f : \mathbb{C}_\infty \longrightarrow \mathbb{C}_\infty \mid f \text{ is continuous}\}.$$

We denote by $\mathcal{R}$ the specific subclass consisting of all non-constant and constant rational functions.

Definition 2.27 (Uniform Metric ρ)**.** The *metric of uniform convergence* $\rho : \mathcal{C} \times \mathcal{C} \longrightarrow \mathbb{R}^+ \cup \{0\}$ on the complex sphere is defined by:

$$\rho(f, g) = \sup_{z \in \mathbb{C}_\infty} \sigma_0(f(z), g(z)),$$

where σ_0 is the spherical metric.

Since the chordal metric σ and the spherical metric σ_0 only differ by a bounded factor, we could equally well use the chordal metric here (the metric would change, but the topology would not).

Now $\mathcal{R}$ is a closed subset of $\mathcal{C}$ because if the rational functions R_n converge uniformly to R on the complex sphere, then R is analytic on the sphere and so it too is rational. More generally (and again we could use either the chordal metric or the spherical metric), we have the following theorem.

Theorem 2.39 (Uniform Limits Preserve Analyticity)**.** *Suppose that each function in a sequence $\{f_n\}$ is analytic in a domain $D \subseteq \mathbb{C}_\infty$, and that f_n converges uniformly on D to f with respect to the spherical metric σ_0. Then f is also analytic in D.*

Proof. See [3, p. 46]. □

Corollary 2.39.1 ($\mathcal{R}$ is Closed)**.** *The space of rational functions $\mathcal{R}$ is a closed subset of the continuous function space $\mathcal{C}$ under the topology induced by the uniform metric ρ.*

With the topology established, we now investigate the topological behavior of the degree function. Crucially, the degree of a rational map cannot change abruptly under small uniform perturbations.

Theorem 2.40 (Continuity of the Degree Function)**.** *The degree map* $\deg : \mathcal{R} \longrightarrow \{0, 1, 2, \dots\}$ *is continuous. In particular, if a sequence of rational functions* R_n *converges uniformly on the Riemann sphere to a function* R*, then* R *is rational and* $\deg(R_n) = \deg(R)$ *for all sufficiently large* n*.*

Proof. See [3, p. 46]. □

Definition 2.28 (Space of Degree n Maps)**.** We denote by $\mathcal{R}_n$ the collection of all rational maps of degree exactly equal to n:

$$\mathcal{R}_n = \{R \in \mathcal{R} \mid \deg(R) = n\}.$$

As a consequence of this theorem (Theorem 2.40), the class $\mathcal{R}_n$ of rational maps of degree n is an open subset of $\mathcal{R}$ for it is the inverse image of the open subset $\{n\}$ of $\mathbb{Z}$ under the continuous map deg. It is easy to see that each $\mathcal{R}_n$ is connected (for if R and S lie in $\mathcal{R}_n$, we can simply "slide" the zeros and poles of R to those of S while maintaining the same degree) so we obtain the following result.

Corollary 2.40.1 (Connected Components of $\mathcal{R}$)**.** *The subspaces* $\mathcal{R}_n$ *are the components of* $\mathcal{R}$*.*

There are a few remarks that I've not included in the notes, because Beardon points out that they can be safely ignored, in [3, p. 47].

Chapter 3

The Fatou and Julia Sets

The informal discussion in Chapter 1 suggested that the dynamics of a rational map R induces a subdivision of the complex sphere into two sets, namely the Fatou set F and the Julia set J of R. Here, we introduce the notion of equicontinuity of a family of maps, and then formally define the Fatou and Julia sets in terms of the equicontinuity of the family $\{R^n\}$. In addition, we consider conditions which imply that a family of analytic maps is equicontinuous (or a normal family) in a domain.

3.1 The Fatou and Julia Sets

Definition 3.1 (Continuous function)**.** Given two metric spaces (X, d_X) and (Y, d_Y). A map $f : X \to Y$ is *continuous at* x_0 in X if and only if, for every positive ε there exists a positive δ such that for all x in X,

$$d_X(x_0, x) < \delta \quad \text{implies} \quad d_Y(f(x_0), f(x)) < \varepsilon.$$

The function f is *continuous on a subset* X_0 *of* X if it is continuous at each point x_0 of X_0.

Clearly, in this definition, δ depends on f, x_0 and ε. For equicontinuity we will look for a δ that works for all f in a family of functions.

Definition 3.2 (Equicontinuous family)**.** Given two metric spaces (X, d_X) and (Y, d_Y). A family $\mathcal{F}$ of maps of (X, d_X) into (Y, d_Y) is *equicontinuous at* x_0 in X if and only if, for every positive ε there exists a positive δ such that for all x in X, and for all f in $\mathcal{F}$,

$$d_X(x_0, x) < \delta \quad \text{implies} \quad d_Y(f(x_0), f(x)) < \varepsilon.$$

The family $\mathcal{F}$ is *equicontinuous on a subset* X_0 *of* X if it is equicontinuous at each point x_0 of X_0.

It is important to realize that the equicontinuity of $\mathcal{F}$ at x_0 is the formal expression for the idea of preservation of proximity which was introduced informally in Chapter 1: indeed, it implies that every function in $\mathcal{F}$ maps the open ball with center x_0 and radius δ into a ball of radius at most ε.

Remark 3.1 (Equicontinuity is preserved under union). If the family $\mathcal{F}$ is equicontinuous on each of the subsets D_α of X, then it is automatically equicontinuous on the union $\cup D_\alpha$.

Taking the collection $\{D_\alpha\}$ to be the class of all open subsets of X on which $\mathcal{F}$ is equicontinuous, this leads to the following general principle.

Theorem 3.2 ($\exists$ a maximal open subset where $\mathcal{F}$ is equicontinuous)**.** *Let* $\mathcal{F}$ *be any family of maps, each mapping* (X, d_X) *into* (Y, d_Y)*. Then there is a maximal open subset of* X *on which* $\mathcal{F}$ *is equicontinuous.*

Corollary 3.2.1 ($\exists$ a maximal open subset where $\{f^n\}$ is equicontinuous)**.** *If* f *maps a metric space* (X, d_X) *into itself, then there is a maximal open subset of* X *on which the family of iterates* $\{f^n \mid n \in \mathbb{N}\}$ *is equicontinuous.*

This (at last) provides us with a formal definition for the Fatou and Julia sets of a rational map R, and the following terminology is chosen to honor the creators of the theory.

Definition 3.3 (Fatou and Julia sets)**.** Let R be a non-constant rational function. The *Fatou set* of R is the maximal open subset of $\mathbb{C}_\infty$ on which $\{R^n \mid n \in \mathbb{N}\}$ is equicontinuous, and the *Julia set* of R is its complement in $\mathbb{C}_\infty$.

We denote the Fatou set of a rational map R by F, and the Julia set by J, although sometimes, when we need to mention R explicitly, we use $F(R)$ and $J(R)$ instead.

Remark 3.3 (F is open and J is compact). Note that, by definition, $F(R)$ is *open*, and $J(R)$ is *compact*.

Although the term "Julia set" is standard, the use of "Fatou set" was suggested as late as 1984 (in [4]). It seems appropriate, but the reader should be familiar with the common alternatives, namely *the stable set*, and *the set of normality*. This latter term is a reference to normal families of analytic functions, but we have preferred to base our definition on equicontinuity because of its more immediate geometric appeal.

We end this section with two simple, but important, properties of F and J. A rational map, and in particular, a Möbius map and its inverse, satisfy a Lipschitz condition with respect to the spherical (and chordal) metric on $\mathbb{C}_\infty$ (2.12). Thus, if we conjugate R with respect to a Möbius map, equicontinuity is transferred in the obvious way, and we have the following theorem.

Theorem 3.4 (Equicontinuity is transferred via conjugation)**.** *Let R be a non-constant rational map, let g be a Möbius map, and let $S = gRg^{-1}$. Then $F(S) = g(F(R))$ and $J(S) = g(J(R))$.*

A similar argument leads to the following theorem.

Theorem 3.5 (F and J are preserved under iteration)**.** *For any non-constant rational map R, and any positive integer p, $F(R^p) = F(R)$ and $J(R^p) = J(R)$.*

Proof. See [3, p. 51]. □

3.2 Completely invariant sets

Definition 3.4 (Invariant set under g)**.** Given a map g of a set X into itself, a subset E of X is:

1. *forward invariant* if $g(E) = E$;
2. *backward invariant* if $g^{-1}(E) = E$;
3. *completely invariant* if $g(E) = E = g^{-1}(E)$.

Remark 3.6 (g surjective $\implies$ backward invariant = completely invariant). If g is surjective (that is, if $g(X) = X$) then the concepts of backward invariance and complete invariance coincide. Indeed, by surjectivity (but not necessarily in the general case, because there may be some points in E where $g^{-1}(x) = \emptyset$),

$$g(g^{-1}(E)) = E$$

which, with backward invariance, yields $g(E) = E$. The requirement of surjectivity here is crucial and without it, there is a difference between backward and complete invariance.

Example 3.1. The map $z \mapsto \exp(z)$ shows how this concepts can be different. Under this map $\mathbb{C}$ is backward invariant, but not completely invariant.

As our main concern is rational maps, we recall that these map $\mathbb{C}_\infty$ onto itself: thus for rational maps, backward invariance coincides with complete invariance.

We illustrate these ideas with the following result which will be of importance later.

Theorem 3.7 (Completely invariant $\implies$ non-empty, non-singleton set)**.** *Let R be a rational map of degree at least two and suppose that a finite set E is completely invariant under R. Then E has at most two elements.*

Proof. See [3, p. 52]. □

It is convenient to collect together several general, but elementary, properties of completely invariant sets and throughout, *we shall assume that* $g : X \to X$ *is surjective* so that we may use complete invariance and backward invariance interchangeably.

Proposition 3.8 (Complete invariance is preserved under conjugation)**.** *Given a surjective map g of a set X into itself, and a completely invariant subset E of X under g. Then, if h is a bijection of X onto itself, then $h(E)$ is completely invariant under hgh^{-1}.*

Proposition 3.9 (Complete invariance is preserved under intersection)**.** *Given a surjective map* g *of a set* X *into itself, and a family of completely invariant subsets* $\{E_\alpha\}$ *of* X *under* g*. Then,*

$$E = \bigcap E_\alpha$$

is completely invariant.

Proof. The proof can be obtained by applying the standard construction which is used to generate, for example, subgroups, subspaces and topologies. The operator g^{-1} commutes with the intersection operator, that is, for any collection $\{E_\alpha\}$ of sets,

$$g^{-1}\left(\bigcap E_\alpha\right) = \bigcap g^{-1}(E_\alpha)$$

□

This means, that we can take any subset E_0 and form the intersection, say E, of all those completely invariant sets which contain E_0: obviously, E is then the *smallest completely invariant set that contains* E_0, and we say that E_0 *generates* E .

We now introduce an equivalence relation which greatly facilitates the discussion of completely invariant sets.

Definition 3.5 (Relation $\sim$ (under g))**.** Given a map g of a set X into itself, for any x and y in X, we define the relation $\sim$ on X by $x \sim y$ if and only if there exists non-negative integers n and m with

$$g^n(x) = g^m(y). \tag{3.1}$$

Remark 3.10 ($\sim$ is an equivalence relation)*.* Obviously the relation $\sim$ is symmetric and reflexive, and it is also transitive for (3.1) and $g^p(y) = g^q(z)$ imply that

$$g^{n+p}(x) = g^{m+q}(z);$$

thus $\sim$ is an equivalence relation on X.

Definition 3.6 (Orbit of x (under g))**.** We denote the equivalence class containing x by $[x]$, and we call this the *orbit* of x (some authors call this the "great orbit").

It is easy to identify the orbit $[x]$, and we have the following theorem.

Theorem 3.11 ($[x]$ is the completely invariant set generated by $\{x\}$)**.** *Let $g : X \to X$ be any map of X onto itself. Then $[x]$ is the completely invariant set generated by the singleton $\{x\}$.*

Proof. See [3, p. 53]. □

Corollary 3.11.1 (Completely invariant $\iff$ union of orbits)**.** *A set E is completely invariant if and only if it is a union of equivalence classes $[x]$.*

This suggests we should also look at the interior, closure and boundary operators.

Theorem 3.12 (Completely invariance is preserved under the complement, interior and boundary operators)**.** *Let g be any continuous, open map of a topological space X onto itself and suppose that E is completely invariant. Then so are the complement $X - E$, the interior E^o, the boundary ∂E and the closure $\overline{E}$, of E.*

Proof. See [3, p. 53]. □

We now prove that the division of the sphere into Fatou and Julia sets is completely invariant: this is fundamental.

Theorem 3.13 (F and J are completely invariant)**.** *Let R be any rational map. Then the Fatou set F and the Julia set J are completely invariant under R.*

Proof. See [3, p. 54]. □

We can say more than Theorem 3.13 when R is a polynomial, and we have the following theorem.

Theorem 3.14 (F_∞ is completely invariant for polynomials)**.** *Let* P *be a polynomial of degree at least two. Then* ∞ *is in* $F(P)$*, and the component* F_∞ *of* F *containing* ∞ *is completely invariant under* P*.*

Proof. See [3, p. 54]. □

We end this section with a simple observation which will be used several times later.

Proposition 3.15 (Periodic invariance of components)**.** *Let* f *be a continuous map of a topological space* X *onto itself, and suppose that* X *has only a finite number of components* X_j*. Then for some integer* m*, each* X_j *is completely invariant under* f^m*.*

Proof. See [3, p. 55]. □

This, of course, was the essential idea in the proof of Theorem 3.7.

3.3 Normal families and equicontinuity

We have defined the Fatou and Julia sets in terms of the equicontinuity of the family $\{R^n\}$. However, equicontinuity is closely related to normal families of analytic functions, and once we have understood this connection (which is expressed in the Arzelà-Ascoli Theorem), we can exploit the more powerful results about normal families of analytic functions to derive further information about the Fatou and Julia sets. This is our program for this section.

We begin with the definitions relevant to normal families.

Definition 3.7 (Pointwise convergence)**.** A sequence (f_n) of maps from a metric space (X, d_X) to a metric space (Y, d_Y) *converges pointwise on* X to some map f if for each x in X: for every positive ε there exists a natural number N such that for all $n \geqslant N$,

$$d_Y(f_n(x), f(x)) < \varepsilon.$$

Definition 3.8 (Uniform convergence)**.** A sequence (f_n) of maps from a metric space (X, d_X) to a metric space (Y, d_Y) *converges uniformly on* X to some map f if for every positive ε there exists a natural number N such that for all x in X, and for all $n \geqslant N$,

$$d_Y(f_n(x), f(x)) < \varepsilon.$$

Remark 3.16 (Uniform convergence $\implies$ pointwise convergence). Note that for a family to be uniformly convergent is a stronger statement than for it to be pointwise convergent. Indeed, if we write the definitions in terms of limits we have:

$$\begin{aligned}
&\lim_{n\to\infty} f_n = f \text{ pointwise} \iff \\
&\qquad\qquad \forall x \in X \lim_{n\to\infty} f_n(x) = f(x); \\
&\lim_{n\to\infty} f_n = f \text{ uniformly} \iff \\
&\qquad\qquad \lim_{n\to\infty} \sup\{d_Y(f_n(x), f(x)) \mid \forall x \in X\} = 0.
\end{aligned}$$

This last expression may also clarify why at the section A Topology on the Rational Functions (2.9), we called ρ the *metric of uniform convergence*. Since, using that metric, convergence in C is the same as uniform convergence.

Definition 3.9 (Local uniform convergence)**.** A sequence (f_n) of maps from a metric space (X, d_X) to a metric space (Y, d_Y) *converges locally uniformly on* X to some map f if each point x of X, has a neighborhood on which f_n converges uniformly to f.

Remark 3.17 (Local uniform convergence $\implies$ uniform convergence in compacts). Given a sequence (f_n) of maps from a uniform metric space (X, d_X) to a metric space (Y, d_Y) that converges locally uniformly on X. Then, it converges uniformly on each compact subset of X.

Definition 3.10 (Normal family)**.** A family $\mathcal{F}$ of maps from a metric space (X, d_X) to a metric space (Y, d_Y) is said to be *normal*, or a *normal* *family*, in X if every infinite sequence of functions from $\mathcal{F}$ contains a subsequence which converges locally uniformly on X_1.

We give the usual reminder that the limit of the convergent subsequence need not lie in $\mathcal{F}$. Although we have formulated these definitions for general metric spaces, we shall only be concerned with subdomains of the complex sphere with the chordal (or spherical) metric, and we come now to a statement of the Arzelà-Ascoli Theorem.

Theorem 3.18 (In $\mathbb{C}_\infty$ with continuous functions: equicontinuous family $\iff$ normal family)**.** *Let D be a subdomain of the complex sphere, and let $\mathcal{F}$ be a family of continuous maps of D into the sphere. Then $\mathcal{F}$ is equicontinuous in D if and only if it is a normal family in D.*

Proof. See [1, p. 222]. □

As an illustration of the direct use of normality rather than equicontinuity, we consider Vitali's Theorem: this exploits analytic continuation and so enables us to derive information about the limit f of a sequence (f_n) throughout a domain D simply from a knowledge of f on only a small part of D.

Theorem 3.19 (Vitali's Theorem)**.** *Suppose that the family $\{f_1, f_2, \ldots\}$ of analytic maps is normal in a domain D, and that f_n converges pointwise to some function f on some non-empty open subset W of D. Then f extends to a function F which is analytic on D, and $f_n \to F$ locally uniformly on D.*

Proof. See [3, p. 56]. □

We turn now to obtain conditions which guarantee the normality of a family of analytic functions.

Proposition 3.20 (In $\mathbb{C}_\infty$ with continuous functions: holomorphic and uniformly bounded $\iff$ equicontinuous family $\iff$ normal family)**.** *Given a family of functions $\mathcal{F}$ which are holomorphic and uniformly bounded in D. Then, $\mathcal{F}$ is equicontinuous, and hence normal in D.*

Proof. It is an elementary consequence of Cauchy's Integral Formula that if f is holomorphic and satisfied $|f| \leqslant M$ in a domain D in $\mathbb{C}$, then $|f'|$ is bounded above on each compact subset K of D, the upper bound depending on M, K, and D *but not on* f. □

It is possible to obtain a variety of mild generalizations of this result without much effort, but all such results are hopelessly inadequate for our purposes. Instead, we pass these by and go straight to one of the most powerful results concerning normality: this result is absolutely essential for the study of iteration of rational maps, yet its substantial proof can safely be omitted without prejudicing an understanding of iteration theory.

Theorem 3.21 (The family of analytic maps $\{f : D \to \mathbb{C}_\infty - \{0, 1, \infty\}\}$ is normal)**.** *Let* D *be a domain on the complex sphere* $\mathbb{C}_\infty$*, and let* Ω *be the domain* $\mathbb{C}_\infty - \{0, 1, \infty\}$*. Then the family* $\mathcal{F}$ *of all analytic maps* $f : D \to \Omega$ *is normal in* D*.*

Proof. See [3, p. 59]. □

The hypothesis of Theorem 3.21 is that each function in $\mathcal{F}$ fails to take the values 0, 1 and ∞ at any point of D. Let's see how Theorem 3.21 can be used to produce two variations on the same theme. In the first one of these, we assume only that each function f fails to take three values which no *may depend on* f. It is too much to expect that three values can be completely arbitrary, and we must add the requirement that they are uniformly separated on the sphere.

Theorem 3.22 (The family of analytic maps $\{f : D \to \mathbb{C}_\infty - \{a_f, b_f, c_f\}\}$ is normal)**.** *Let* $\mathcal{F}$ *be a family of maps, each analytic in a domain* D *on the complex sphere. Suppose also that there is a positive constant* m *and, for each* f *in* $\mathcal{F}$*, three distinct points* a_f*,* b_f *and* c_f *in* $\mathbb{C}_\infty$ *such that:*

1. f *in* $\mathcal{F}$ *does not take the values* a_f*,* b_f *and* c_f *in* D*; and*
2. $\min\{\sigma(a_f, b_f), \sigma(b_f, c_f), \sigma(c_f, a_f)\} \geqslant m$.

Then $\mathcal{F}$ *is normal in* D*.*

Proof. See [3, p. 58]. □

Of course, Theorem 3.22 contains Theorem 3.21 as a special case, for we can take $a_f = 0$, $b_f = 1$ and $c_f = \infty$ for every f. There is another variation of Theorem 3.21 that we shall need.

Theorem 3.23 (The family of analytic maps $\{f : D \to \mathbb{C}_\infty\}$ such that $f(z) \neq \varphi_j(z)$ for $j = 1, 2, 3$, is normal)**.** *Let D be a domain, and suppose that the functions φ_1, φ_2 and φ_3 are analytic in D, and are such that the closures of the domains $\varphi_j(D)$ are mutually disjoint. Now let $\mathcal{F}$ be a family of functions, each analytic in D, such that for every z in D, and every f in $\mathcal{F}$, $f(z) \neq \varphi_j(z)$, $j = 1, 2, 3$. Then $\mathcal{F}$ is normal in D.*

Proof. See [3, p. 58]. □

Chapter 4

Properties of the Julia Set

This chapter is devoted to developing the basic and most easily available properties of the Julia set of a rational map. Other properties of the Julia set occur throughout the text.

4.1 Exceptional Points

We begin now to exploit the fundamental Theorem 3.21. Given a rational map R, we have the induced equivalence relation $\sim$ discussed in section 3.2, and the equivalence class $[z]$ is the smallest completely invariant set which contains z. It is easy to see that $[z]$ can be finite only in rather special circumstances, so we enshrine this idea in a formal definition.

Definition 4.1 (Exceptional point)**.** A point z is said to be *exceptional* for R when $[z]$ is finite, and the set of such points is denoted by $\mathsf{E}(\mathsf{R})$.

First, we justify the terminology by showing that such points are indeed exceptional.

Theorem 4.1 (Rational maps have at most two exceptional points)**.** *A rational map R of degree at least two has at most two exceptional points.*

1. *If $\mathsf{E}(\mathsf{R}) = \{\zeta\}$, then R is conjugate to a polynomial with ζ corresponding to ∞.*

2. *If $\mathsf{E}(\mathsf{R}) = \{\zeta_1, \zeta_2\}$, then R is conjugate to some map $z \mapsto z^{\mathsf{d}}$, where d is an integer ($\mathsf{d} \in \mathbb{Z}$), and ζ_1 and ζ_2 correspond to 0 and ∞.*

Proof. See [3, p. 66]. □

From this and Theorems 3.5 and 3.14, we have immediately the following result.

Corollary 4.1.1 (Exceptional points of R with $\deg(R) \geqslant 2$ lie in F)**.** *If* $\deg(R) \geqslant 2$*, then the exceptional points of* R *lie in* $F(R)$*.*

We turn now to another characterization of exceptional points.

Definition 4.2 (Backward orbit and predecessors)**.** For any z, the *backward orbit* of z is the set

$$\begin{aligned} \mathcal{O}^-(z) &= \{w \mid \text{for some } n \geqslant 0, R^n(w) = z\} \\ &= \bigcup_{n \geqslant 0} R^{-n}\{z\}, \end{aligned}$$

and we call the points in $\mathcal{O}^-(z)$ the *predecessors* of z.

Remark 4.2 (Exceptional $\Longrightarrow$ finite backward orbit). As $\mathcal{O}^-(z) \subset [z]$, any exceptional point z has a finite backward orbit.

We show now that the converse is true.

Theorem 4.3 (Characterization of exceptional points)**.** *The backward orbit* $\mathcal{O}^-(z)$ *is finite if and only if* z *is exceptional.*

Proof. See [3, p. 66]. □

As an application of this result we can obtain the following theorem.

Theorem 4.4 (R^k is conjugate to a polynomial $\Longrightarrow$ R^2 is conjugate to a polynomial)**.** *If for some positive integer* k*,* R^k *is conjugate to a polynomial, then* R^2 *is also conjugate to a polynomial.*

Proof. Suppose that some iterate R^k of R is conjugate to a polynomial. Then there is some w with $\left(R^k\right)^{-1}\{w\} = \{w\}$ (see Theorem 2.19) and using Theorem 4.3 this implies that w is exceptional for R. Now such map need not be conjugate to a polynomial (consider, for example, $z \mapsto z^{-2}$) but from Theorem 4.1, R^2 must be. Thus, we have proved the theorem. ☐

4.2 Properties of the Julia Set

We shall continue to exploit Theorem 3.21 as we explore the structure of the Julia set. We have not yet considered whether the Julia set can be non-0empty, and our first task will be to settle this issue. The case when the rational map R has degree one is trivial and of little interest, but in all other cases, J is not empty.

Theorem 4.5 ($\deg(R) \geqslant 2 \implies J$ is infinite)**.** *If* $\deg(R) \geqslant 2$*, then* $J(R)$ *is infinite.*

Proof. See [3, p. 67]. ☐

It is clear from Theorem 3.21 that the integer three has a significant role to play, and it appears yet again in a decisive way in the next result.

Theorem 4.6 (J is minimal)**.** *Let* R *be a rational map with* $\deg(R) \geqslant 2$*, and suppose that* E *is a closed, completely invariant subset of the complex sphere. Then either:*

1. E *has at most two elements and* $E \subset E(R) \subset F(R)$*; or*

2. E *is infinite and* $E \supset J(R)$*.*

Proof. See [3, p. 68]. ☐

It is usual, and convenient, to express the conclusion of Theorem 4.6 in the form: J *is the smallest, closed, completely invariant set with at least three points*, and we shall refer to this property as the *minimality* of J. The proof of Theorem 4.6 seems surprisingly short in view of the interesting corollaries which follow from it but, of course, this is merely a reflection of the potency of Theorem 3.21. We now begin to develop these corollaries.

Remark 4.7 ($\mathbb{C}_\infty = J^o \cup \partial J \cup F$). First, the extended plane $\mathbb{C}_\infty$ is the disjoint union of the interior J^o of J, the boundary ∂J of J, and the Fatou set F.

Theorem 4.8 ($J = \mathbb{C}_\infty$ or $J^o = \emptyset$). *Either* $J = \mathbb{C}_\infty$ *or* J *has empty interior.*

Proof. We know that F is completely invariant: likewise, J is and hence so are J^o and ∂J (Theorem 3.12). If F is not empty then $F \cup \partial J$ is an infinite, closed, completely invariant set and so, by the minimality of J, it contains J. In these circumstances, $J \subset \partial J$ and this yields the result. □

We shall consider the case when $J = \mathbb{C}_\infty$ in the next section.

Before examining the structure of the Julia set J, let us recall a fundamental definition from topology. Intuitively, a perfect set is one that is solid, containing no gaps and no lonely, isolated points.

Definition 4.3 (Perfect set). A set S is called **perfect** if it is closed and contains no isolated points. Equivalently, S is perfect if it equals its own derived set, meaning $S = S_0$ (where S_0 is the set of all accumulation points of S).

Next, we show that the Julia set fits this description perfectly by analyzing its behavior under the rational map R.

Proposition 4.9 (J is a perfect set). *The derived set* J_0 *of* J *is an infinite, closed, and completely invariant subset of* J. *By the minimality of* J, *it follows that* $J \subset J_0$, *which implies* $J = J_0$. *Consequently,* J is a perfect set (and thus has no isolated points).

Proof. First, since J is infinite (by Theorem 4.5), it must contain accumulation points, so J_0 is non-empty. By definition, any derived set is automatically closed.

Next, we check invariance under the rational map R (which is continuous and of finite degree):

- Since R is continuous, it maps accumulation points to accumulation points, so $R(J_0) \subset J_0$, which means $J_0 \subset R^{-1}(J_0)$ (forward invariance).
- Since R is an open map, the preimage of an accumulation point is also an accumulation point, giving us $R^{-1}(J_0) \subset J_0$ (backward

invariance).

Combining these, we see that J_0 is completely invariant. Finally, by the minimality of the Julia set (Theorem 4.6), any such set within J must be dense and infinite, verifying that $J = J_0$. By our definition, J is perfect. □

Theorem 4.10 (J is uncountable)**.** *Let* R *be a rational map with* $\deg(R) \geqslant 2$*. Then the Julia set* J *is uncountable.*

Proof. By our previous proposition, the Julia set J is a nonempty, closed, and complete metric space that contains no isolated points.

To see why J must be uncountable, suppose for contradiction that J is countable. This means we can list its points as a sequence:

$$J = \{x_1, x_2, x_3, \dots\} = \bigcup_{i=1}^{\infty} \{x_i\}$$

Let $E_i = \{x_i\}$ be the singleton set containing only the point x_i. We analyze its topological properties within J:

1. Each singleton $\{x_i\}$ is closed because finite points are closed in metric spaces.

2. Because J has no isolated points, no single point can form an open neighborhood by itself. Therefore, the interior of each $\{x_i\}$ is completely empty.

A closed set with an empty interior is, by definition, *nowhere dense*.

Thus, we have written J as a countable union of nowhere dense sets $\bigcup_{i=1}^{\infty} E_i$. In topology, such a union is called a set of the *first category*.

However, Baire's Category Theorem states that no nonempty complete metric space can be written as a countable union of nowhere dense sets. This directly contradicts our setup.

Because this contradiction arises solely from assuming J is countable, we conclude that J must be uncountable. □

We continue to explore the consequences of Theorem 3.21, and the next result says that the successive iterates of R increasingly expand a neighborhood of any point J (we have seen this phenomenon before in some of the examples studied in Chapter 1). Given an open set W which meets J, the result describes the extent to which the images $R^n(W)$ cover the sphere. We recall $E(R)$ has at most two elements.

Theorem 4.11. *Let R be a rational map of degree at least two, and let W be any non-empty open set which meets J. Then:*

1. $\cup_{n=0}^{\infty} R^n(W) \supset \mathbb{C}_\infty - E(R)$; *and*
2. *for all sufficiently large integers n, $R^n(W) \supset J$.*

Proof. See [3, p. 69]. □

The next three results show how the Julia set J is realized as the limit, or the closure, of some countable set.

Definition 4.4 (Periodic point)**.** A point ζ is a *periodic point* of R if it is fixed by some iterate R^n.

Remark 4.12 (The number of periodic points is countable). There are only a countable number of periodic points.

Theorem 4.13 ($J \subset \overline{\{\text{periodic points}\}}$)**.** *Suppose that* $\deg(R) \geqslant 2$. *Then J is contained in the closure of the set of periodic points of R.*

Proof. See [3, pp. 70–71]. □

Remark 4.14 ($J \subset \overline{[z]}$). We now consider the relationship between J and an orbit $[z]$ of a non-exceptional point z. As the closure $\overline{[z]}$ of $[z]$ is both closed and completely invariant, the minimality of J implies that *if z is not exceptional, then* $J \subset \overline{[z]}$.

Remark 4.15 ($z \in J \implies J = \overline{[z]}$). If z is in J, then it is not exceptional, so $[z]$, and hence also $\overline{[z]}$, is contained in J: thus *if z is in J, then* $J = \overline{[z]}$.

In fact, we can replace $[z]$ by the smaller set $O^-(z)$ and still prove the following theorem.

Theorem 4.16 ($J \subset \overline{\mathcal{O}^{-}(z)}$ and if $z \in J$ they are equal)**.** *Let* R *be a rational map with* $\deg(R) \geqslant 2$.

1. *If* z *is not exceptional, then* J *is contained in the closure of* $\mathcal{O}^{-}(z)$.
2. *If* $z \in J$*, then* J *is the closure of* $\mathcal{O}^{-}(z)$.

Proof. See [3, p. 71]. □

Remark 4.17 (Using a computer to illustrate J). This shows that for any non-exceptional z the backward orbit $\mathcal{O}^{-}(z)$ is a countable set which accumulates at every point of J, and this is often used as the basis of a computer program to illustrate J. Roughly speaking, one select a non-exceptional z and then computes successive inverse images of z to plot the set J where these accumulate. There are difficulties, however; for example, if $\deg(R) = d$, then the number of inverse images at the n-th stage is d^n and this grows too rapidly for convenient computations, so some choices need to be made (see [8] and [2] for more details).

If z is in F (and is non-exceptional), then the closure of $\mathcal{O}^{-}(z)$ contains J but, of course, is strictly larger than J. Instead of looking at the entire inverse orbit $\mathcal{O}^{-}(z)$ we can also look at the n-th inverse image $R^{-n}(z)$ (which contains at most d^n points) and ask when (in some sense) do these points converge to J? The same question can be asked of a set E instead of the singleton $\{z\}$, and this is a more useful result to have available.

Theorem 4.18 ($R^{-n}(E)$ converges to J for E compact and non-F-limit)**.** *Let* R *be a rational map of degree at least two, and let* E *be a compact subset of the complex sphere with the property that for all* z *in* $F(R)$*, the sequence* $\{R^n(z) \mid n \geqslant 1\}$ *does not accumulate at any point of* E*. Then given any open set* U *which contains* $J(R)$, $R^{-n}(E) \subset U$ *for all sufficiently large* n.

Proof. See [3, p. 72]. □

Remark 4.19. If the conclusion of the theorem holds we say that $R^{-n}(E)$ converges to J and write $R^{-n}(E) \to J$. Examples of such sets E are easy to find: for example, if F_0 is a component of the Fatou set that contains an attracting fixed point ζ, then any compact subset of $F_0 - \{\zeta\}$ has the required property.

We end this section with the following generalization of Theorem 3.5.

Theorem 4.20 (J is preserved between commuting functions)**.** *Suppose that* R *and* S *are rational maps, each of degree at least two. If* R *and* S *commute, then* $J(R) = J(S)$.

Proof. See [3, pp. 72–73]. □

4.3 Rational Maps with Empty Fatou Set

Example 4.1. Our first objective is to show that for the rational map

$$z \mapsto \frac{(z^2+1)^2}{4z(z^2-1)}, \tag{4.1}$$

the Julia set is the entire complex sphere: this example (the first of its kind was given by Lattès in 1918, [7]). After this, we shall make some general observations about other rational functions with this property.

We shall use the technique illustrated in section 1.4, although here, we use a group of translations with two generator, and we require considerably more complex analysis. Let λ and μ be complex numbers that are not real multiples of each other, and let Λ be the corresponding lattice, that is,

$$\Lambda = \{m\lambda + n\mu \mid n, m \in \mathbb{Z}\}. \tag{4.2}$$

A *period parallelogram* for Λ is any closed parallelogram of the form

$$\Omega = \{z + s\lambda + t\mu \mid 0 \leqslant s \leqslant 1, 0 \leqslant t \leqslant 1\}.$$

A non-constant function f is an *elliptic function* for Λ if it is meromorphic on $\mathbb{C}$, and if each ω in Λ is a period of f; that is, if for all z in $\mathbb{C}$ and all ω in Λ, $f(z+\omega) = f(z)$. Of course, any such function maps $\mathbb{C}$ into $\mathbb{C}_\infty$ and, because $f(\mathbb{C}) = f(\Omega)$, we see that $f(\mathbb{C})$ is a compact subset of $\mathbb{C}_\infty$. However, by the Open Mapping Theorem, $f(\mathbb{C})$ is also an open subset of $\mathbb{C}_\infty$: thus

$$f(\Omega) = f(\mathbb{C}) = \mathbb{C}_\infty.$$

Our argument is based on the properties of the Weierstrass elliptic function

$$\wp(z) = \frac{1}{z^2} + \sum{}^{*} \left[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} \right]$$

where $\sum{}^{*}$ denotes summation over non-zero elements ω in Λ (see for example, [1, pp. 272–276], [5] or [6]). We shall only give the main steps in the argument but, for the convenience of some readers, we give and expanded version in the Appendix to this chapter. Now $\wp$ and its derivatives satisfy certain algebraic identities, and among these there is the *addition formula*

$$\wp(2z) = R(\wp(z)), \tag{4.3}$$

where r is the rational function given by

$$R(z) = \frac{z^4 + g_2 z^2/2 + 2g_3 z + (g_2/4)^2}{4z^3 - g_2 z - g_3}, \tag{4.4}$$

and where g_2 and g_3 are known quantities defined in terms of the lattice Λ. Note that (4.3) is analogous to the double angle formulae in elementary trigonometry.

Now let D be any disc in $\mathbb{C}$, let $U = \wp^{-1}(D)$, and define $\varphi(z) = 2z$. As U is open, and as $\varphi^n(U)$ is the set U expanded by a factor 2^n, it is clear that for sufficiently large n, $\varphi^n(U)$ contains a period parallelogram Ω of $\wp$. Using this with (4.3), we deduce that for these n,

$$R^n(D) = R^n(\wp(U)) = \wp(2^n U) = \mathbb{C}_\infty :$$

roughly speaking, this implies that the functions R^n explode any small disc D onto the whole sphere. Clearly, as D is arbitrary, (4.4) implies that the family $\{R^n\}$ is not equicontinuous on *any* open subset of the complex sphere, and so we deduce that $J(R) = \mathbb{C}_\infty$.

This argument gives a family of rational maps whose Julia set is the sphere; indeed, we always have

$$g_2^3 - 27g_3^2 \neq 0, \tag{4.5}$$

and any pair (g_2, g_3) satisfying this is realized by some lattice (see [6, p. 39]). Using this, we can construct Λ so that $g_2 = 4$ and $g_3 = 0$; then R given by (4.4) is the function (4.1). However, there is an easier construction than this. First, we let τ be a positive number, and we construct the square lattice Λ, and the corresponding function $\wp$, by taking $\lambda = \tau$

and $\mu = i\tau$. This choice leads easily to $g_3 = 0$ (see the Appendix in [3, pp. 77–79]), and hence to a simplification of (4.4), namely

$$R(z) = \frac{16z^4 + 8g_2z^2 + (g_2)^2}{16z(4z^2 - g_2)}.$$

Now from (4.5), as $g_3 = 0$, we have $g_2 \neq 0$, so by taking $h(z) = 2z/\sqrt{g_2}$, we find that hRh^{-1} (which also has $\mathbb{C}_\infty$ as its Julia set) is the function given in (4.1). For further information and related examples, see references in [3, p. 75].

There are other interesting dynamical aspects of this example and briefly, we shall discuss some of these. For each positive integer k, we denote the set of points $\omega/2^k$, $\omega \in \Lambda$, by $2^{-k}\Lambda$, and observe that if z is in $2^{-k}\Lambda$, then

$$R^k(\wp(z)) = \wp(2^kz) = \infty.$$

This shows that the inverse orbit $O^-(\infty)$ under R is $\wp(\cup_{k\in\mathbb{N}}2^{-k}\Lambda)$. As $\cup_{k\in\mathbb{N}}2^{-k}\Lambda$ is dense in $\mathbb{C}$, its $\wp$-image is dense in $\mathbb{C}_\infty$, and so by Theorem 4.16, we find again that $J(R) = \mathbb{C}_\infty$.

Next, the formula (4.3) shows that if $m = 2^k - 1$, then

$$R^k(\wp(\omega/m)) = \wp(2^k\omega/m) = \wp(\omega + \omega/m) = \wp(\omega/m),$$

and so the points $\wp(\omega/m)$ are periodic points of R. By considering all ω and all k, we now see that *the periodic points of R are dense in $\mathbb{C}_\infty$.* Although we are not yet in a position to claim that this shows that $J(R) = \mathbb{C}_\infty$, it is intuitively clear that it must (and it will follow from some of our later results).

Definition 4.5 (Pre-periodic point)**.** A critical point of a rational map R is said to be *pre-periodic* if it is not periodic, but a forward image of it is.

Theorem 4.21 ($\forall z \in C$, z pre-periodic $\implies J = \mathbb{C}_\infty$)**.** *If every critical point of the rational map R is pre-periodic, then $J = \mathbb{C}_\infty$.*

Proof. See Chapter 9 of [3]. □

Example (Continuation)**.** The rational map R in (4.4) is of degree four, so it has six critical points, say $c_1, \dots, c_6$. We shall show now that each c_j is finite, and that each is mapped to ∞ (a fixed point of R) by R^2: then each c_j is pre-periodic and we have $J = \mathbb{C}_\infty$. Again, we only sketch the main ideas and refer the reader to the Appendix for more details.

We shall now find the critical points of R in (4.4). Let $\zeta_1, \dots, \zeta_6$ be the six points

$$\lambda/4, \quad \mu/4, \quad (\lambda+\mu)/4, \quad (3\lambda+\mu)/4, \quad (2\lambda+\mu)/4, \quad (3\lambda+2\mu)/4,$$

and let $\eta_1, \dots, \eta_6$ be the six points

$$3\lambda/4, \quad 3\mu/4, \quad (3\lambda+3\mu)/4, \quad (\lambda+3\mu)/4, \quad (2\lambda+3\mu)/4, \quad (\lambda+2\mu)/4:$$

note that these twelve points are distinct. As $\wp$ is an even function, and as $\eta_j + \zeta_j \in \Lambda$, we find that

$$\wp(\eta_j) = \wp(-\zeta_j) = \wp(\zeta_j),$$

and from this, and the fact that modulo Λ, each point of $\mathbb{C}_\infty$ has precisely two pre-images under $\wp$, we see that the six values $\wp(\zeta_j)$ are distinct. Now $\wp'(z) = \infty$ if and only if $\wp(z) = \infty$, that is, if and only if z is in Λ, and it is also known that $\wp'(z) = 0$ if and only if z is in $2^{-1}\Lambda$ but not in Λ. It follows that

$$\wp'(\zeta_j) \neq 0, \infty, \quad \wp'(2\zeta_j) = 0,$$

and as

$$R'(\wp(z))\wp'(z) = 2\wp'(2z),$$

we see that the six values $\wp(\zeta_j)$ are the six critical points of R. Finally, it is immediate from (4.3) that R^2 maps each of these to ∞.

Example 4.2. Theorem 4.21 enables us to produce other examples of maps whose Julia set is the sphere: for example, the map

$$R(z) = \frac{(z-2)^2}{z^2}$$

has critical points at 0 and 2, and as $2 \to 0 \to \infty \to 1 \to 1$ under an application of R, Theorem 4.21 implies that for this function too, $J(R)$ is the complex sphere. It is easy to see that this R is conjugate to the map

$$z \mapsto 1 - 2/z^2$$

and in fact, many functions in the family $1 + \omega/z^2$ have this same property: see reference in [3, p. 76] for further details.

We end this section with a characterization of those rational maps whose Julia set is the entire sphere, reference [53] in [3].

Theorem 4.22 ($J = \mathbb{C}_\infty \iff \exists z$ with dense forward orbit)**.** *Let* R *be a rational map. Then* $J(R) = \mathbb{C}_\infty$ *if and only if there is some* z *whose forward orbit* $\{R^n(z) \mid n \geqslant 1\}$ *is dense in the complex sphere.*

Proof. See [3, pp. 76–77]. □

Chapter 5

The Structure of the Fatou Set

We begin with the simple properties of the Fatou set of a rational map R. As R is a branched convering map of the sphere onto itself with the critical points as branch points, the position of the critical points influences the structure of the Fatou set F through the Euler characteristic. We discuss these ideas and apply them to study the structure of F.

5.1 The Topology of the Sphere

Notation

Basic Math:

- fg: Composition of f and g ($f \circ g$).
- f^n: The composition of f with itself n times.
- $I = f^0$: Identity function.
- $f^{(n)}$: n-th derivative of f.
- $\sim$: Equivalence relation.
- $[x]$: Equivalence class of x (in some contexts orbit of x).
- $\mathbb{N}$: Naturals without 0.
- $\mathbb{N}_0$: Naturals with 0.

Complex Analysis:

- $\mathbb{C}_\infty$: Extended complex plane.
- Δ: Unit disk in $\mathbb{C}$.
- $R(z)$: Rational function in complex variable z.
- σ: Chordal metric.
- σ_0: Spheric metric.
- γ: A curve.
- $E(\gamma)$: Exterior component of a closed curve.
- $v_f(z_0)$: Valency of f at z_0.

Topology:

- N: Neighborhood of a point.
- E^o: Interior of a set.
- $\overline{E}$: Closure of a set.
- ∂E: Boundary of a set.

Complex Dynamics:

- ζ: Fixed or periodic point.
- $\deg(R)$: Degree of R.
- $m(R, z_0)$: Multiplier of R at z_0.
- F or $F(R)$: Fatou set (of R).
- J or $J(R)$: Julia set (of R).
- C or $C(R)$: Set of critical points (of R).
- V or $V(R)$: Set of critical values (of R).
- $E(R)$: Set of exceptional points of R.
- $\mathcal{O}(z)$ or $[z]$: Orbit of z.
- $\mathcal{O}^-(z)$: Backward orbit of z.

Bibliography

[1] Lars Ahlfors. *Complex Analysis: An Introduction to the Theory of Analytic Functions of One Complex Variable*. 3rd ed. International Series in Pure & Applied Mathematics. Mcgraw Hill Book Co, 1979. ISBN: 9780070006577.

[2] Michael F. Barnsley et al. *The Science of Fractal Images*. Ed. by Heinz-Otto Peitgen and Dietmar Saupe. Springer New York, 1988. ISBN: 9781461237846. DOI: 10.1007/978-1-4612-3784-6.

[3] A.F. Beardon. *Iteration of Rational Functions: Complex Analytic Dynamical Systems*. Graduate Texts in Mathematics. Springer New York, 1991. ISBN: 9783540975892.

[4] Paul Blanchard. "Complex analytic dynamics on the Riemann sphere". In: *Bulletin of the American Mathematical Society* 11.1 (1984), pp. 85–141. ISSN: 0273-0979. DOI: 10.1090/s0273-0979-1984-15240-6.

[5] Patrick Du Val. *Elliptic Functions and Elliptic Curves*. London Mathematical Society Lecture Note Series. Cambridge University Press, Aug. 1973. ISBN: 9781107359901. DOI: 10.1017/cbo9781107359901.

[6] Serge Lang. *Elliptic Functions*. Springer New York, 1987. ISBN: 9781461247524. DOI: 10.1007/978-1-4612-4752-4.

[7] Samuel Lattes. "Sur l'iteration des substitutions rationelles et les fonctions de Poincaré". In: *CR Acad. Sci. Paris* 166.3 (1918), pp. 26–28.

[8] Heinz-Otto Peitgen and Peter H. Richter. *The Beauty of Fractals*. Springer Berlin Heidelberg, 1986. ISBN: 9783642617171. DOI: 10.1007/978-3-642-61717-1.